MILITARY SPECIFICATION

CONTROL ASSEMBLY, GYROSCOPE, ATTITUDE-HEADING A/A24G-42

1. SCOPE

1.1 This specification covers the requirements for an attitude-heading, gyroscope control assembly, designated A/A24G-42.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connectors, Electric, "AN" Type
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification for
MIL-G-25591	Gyroscope, Rate Switching, Type MC-1
MIL-C-25773	Compass System, Gyroscopic AN/AJN-3A
MIL-D-26503	Detector, Magnetic, Azimuth, and Compensator, Detector, Magnetic Azimuth
MIL-I-27193	Indicator, Attitude ARU-2B/A
MIL-C-27205	Control Assembly, Gyroscope, Attitude-Heading A/A24G-5A
MIL-B-27497	Bearings, Jewel, Sapphire or Ruby, Synthetic
MIL-I-27710	Indicator, Attitude, Remote

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property

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MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Electronic Equipment
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-704	Electric Power, Aircraft, Characteristics and Utilization of
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3113	Connector, Receptacle, Electric Solder Type, Solder Mounting, Bayonet Coupling
MS17322	Meter, Time Totalizing Miniature, 400 Cycle, Digital

(Copies of documents required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Northrop Specifications

14-62903	Sight System, Lead Computing Optical, dated 1 June 1970
14-62909	Fire Control Radar, Air to Air, dated 8 June 1970

(Application for copies should be addressed to the Northrop Corporation, Aircraft Division, 3901 West Broadway, Hawthorne, California 90250 or to the procuring activity.)

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components. The control assembly shall consist of the following components:

a. SBK-16/A24G-42 displacement gyroscope

b. ASK-8A/A24G-5A electronic control amplifier.

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3.2.1 Configuration. The control assembly shall supply one synchro output for roll, pitch, and heading, one pitch linear potentiometer output, and four roll sine-cosine potentiometer outputs.

3.2.1.1 Electrical wiring. The control assembly shall be wired in accordance with tables I and II.

3.2.1.2 Interface. The control assembly shall be compatible with and shall meet all requirements of this specification when operated in conjunction with the following associated equipment. Such associated equipment will not be supplied as a part of the control assemblies. The control assembly shall not be damaged by operation in any mode of the associated equipment nor shall it be damaged in any mode of operation when any or all of the associated equipment is partly or wholly disconnected from it. The performance of the associated equipment shall in no way be degraded or interfered with by operation of the control assembly.

a. ARU-2B/A Attitude Director Indicator (MIL-I-27193)

b. ARU-20/A Attitude Indicator, Remote (MIL-I-27710)

c. DT-309/AJN Magnetic Azimuth Detector (MIL-D-26503)

d. Type MC-1 Rate Switching Gyroscope (MIL-G-25591).

3.2.1.2.1 Gyroscope. The displacement gyroscope shall be capable of meeting all requirements specified herein when interfaced with the equipment specified in 3.2.1.2 and the following:

a. Fire Control Radar, Air to Air (Northrop Specification 14-62909)

b. Sight System, Lead Computing, Optical (Northrop Specification 14-62903).

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practicable to avoid them. Where used and not hermetically sealed, they shall be treated with a fungicidal agent acceptable to the procuring activity. However, if they are to be used in a hermetically sealed inclosure, fungicidal treatment will not be necessary.

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TABLE I. DISPLACEMENT GYROSCOPE CONNECTOR J9

Pin	Connector J9-MS3113H18C32PW, or equal
A	Neutral
B	AØ Input power - 3-phase 400 Hz 200V
C	BØ Line-to-line, 115V line-to-neutral
D	CØ A-B-C phase rotation, Y connected
E	Spare
F	Azimuth transmitter synchro No. 2 rotor for excitation with 115V 400 Hz
G	Signal ground
H	Pitch erection (when connected to pin S)
J	Common to power neutral
K	Leveling (when connected to pin Y)
L	Fast erection (when connected to pin J)
M	+15V dc
N	-15V dc
P	D-C return
R	Bank erection (when connected to pin S)
S	Erection common
T	Precession torquing - pitch
U	Precession torquing - bank
V	Spare
W	Power failure warning
X	
Y	Excitation to leveling torquer
Z	$+P_1$ excitation
a	$-P_1$ excitation
b	$+P_2$ excitation
c	$-P_2$ excitation
d	$+P_3$ excitation
e	$-P_3$ excitation
f	P_3 cosine roll
g	P_3 cosine roll return
h	15 cosine roll
i	15 cosine roll return

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TABLE II. DISPLACEMENT GYROSCOPE CONNECTOR J-8

Pin	Connector J8-MS3113H18C30PW, or equal
A(X) B(Z) C(Y)	Inner roll transmitter synchro stator (internally excited rotor, neutral to CØ)
D(Z) E(Y) F(X)	Pitch transmitter synchro stator (internally excited rotor, neutral to CØ)
G(X) H(Z) J(Y)	Roll transmitter synchro stator (internally excited rotor, neutral to CØ)
K(Z) L(X) M(Y)	Azimuth transmitter synchro stator No. 1 (internally excited rotor, neutral to CØ)
N(Z) P(Y) R(X)	Azimuth transmitter synchro stator No. 2 (externally excited rotor)
S T U V W	Azimuth slaving torquer
X	P_1 Sine roll
Y	P_1 Cosine roll
Z	P_2 Sine roll
a	P_2 Cosine roll
b	P_3 Sine roll
c	P_3 Sine roll return
d	15 Sine roll
e	15 Sine roll return
f	Pitch output
g	Spare

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3.4.2 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion under all conditions specified herein and those likely to be met in storage or normal service.

3.4.2.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion by means of protective coating or hermetic sealing, intimate contact of dissimilar metals shall be avoided. Dissimilar metals are defined in MIL-STD-889.

3.4.3 Nonmagnetic materials. Nonmagnetic materials shall be used for all par of the control assembly except where magnetic materials are essential.

3.4.4 Corrosive fumes. The materials, as installed in the assembly and under the service conditions specified herein, shall not liberate deleterious fumes.

3.4.5 Protective treatment. When materials are used in the construction of the control assembly that are subject to deterioration when exposed to climati and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prever compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The control assembly shall be so designed and constructed as to meet the requirements specified herein. It shall withstand the strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. The control assembly shall be of such design as to facilitate ease of maintenance.

3.5.1 Maintainability. The contractor shall establish a maintainability mean down-time value based on an analysis of his basic design.

3.5.2 Reliability. The control assembly shall be so designed that each component will have a minimum acceptable mean-time-between-failures (MTBF) of 570 hours based on a 90-percent confidence level.

3.5.3 Displacement gyroscope. The displacement gyroscope shall consist of a vertical gyro and a directional gyro mounted in multiple gimbals with a common outer gimbal as specified herein. This gimbal arrangement is used to avoid the gimbal-lock position.

3.5.3.1 Vertical gyro. The vertical gyro shall have sufficient angular momentum and be of such design as to provide the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges specified.

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3.5.3.1.1 Gimbal system. The vertical gyro shall be suspended in a gimbal system which has two free inner gimbals and a servo-driven outer gimbal to eliminate gimbal lock of the gyro. The outer gimbal shall be used in common with the directional gyro.

3.5.3.1.1.1 First vertical gyro gimbal. The first gimbal shall be the gyro rotor housing. It shall be stabilized in roll by the inertia of the gyro.

3.5.3.1.1.2 Second vertical gyro gimbal. The second gimbal shall be the next gimbal outward from the first gimbal. It shall be stabilized in pitch by the gyro.

3.5.3.1.1.3 Third vertical gyro gimbal. The third gimbal shall be the outer gimbal. It shall be servo-driven to maintain a fixed relation between the first and second gimbals except for certain conditions specified herein.

3.5.3.1.2 Vertical gyro axes of freedom. The vertical gyro shall have three axes of freedom. Each shall be a signal axis.

3.5.3.1.2.1 First vertical gyro axis of freedom. The first axis of freedom shall be between the first and second gimbals. This axis shall be maintained in the horizontal plane and shall be aligned with the orthogonal horizontal projection of the fore-and-aft axis of the reference assembly under normal conditions. A minimum freedom of $\pm 70^{\circ}$ shall be provided about this axis. An angular displacement about the axis shall provide an indication which shall change from roll to yaw as the pitch angle changes from 0° pitch to $\pm 90^{\circ}$ pitch. Stops shall be incorporated on the axis to produce a pitch precession when the freedom is exceeded; no damage shall result under this condition.

3.5.3.1.2.2 Second vertical gyro axis of freedom. The second axis of freedom shall be between the second and third (servo-driven) gimbals. This axis shall be perpendicular to the first axis of freedom and shall be maintained in the horizontal plane under normal conditions. Unlimited freedom shall be provided about the axis. An angular displacement about the axis shall be a measurement of the pitch angle.

3.5.3.1.2.3 Third vertical gyro axis of freedom. The third axis of freedom shall be between the third gimbal and the instrument frame or case. This axis shall be perpendicular to the second axis of freedom and shall coincide with the fore-and-aft axis of the instrument. It shall be positioned by a pickoff on the bank axis of the vertical gyro. Unlimited freedom shall be a characteristic of the roll angle.

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3.5.3.1.3 Gimbal actuation. The inner two gimbals shall be maintained in a fixed relation to the apparent gravity vertical by the inertia of the gyro. The outer gimbal shall be activated by a servo mechanism and so driven as to maintain the spin axis of the vertical gyro and the pitch axis in a 90° relation to each other, except for certain transient conditions. The response of the servo of the outer gimbal shall be such that this 90° relation will be maintained during roll rates from 0° through 360° per second with no lag, overshooting, oscillation, hunting, or jumping of such magnitude and frequency as to affect the performance of a high-performance autopilot. Both roll and pitch drives shall originate from the instrument frame. The servo loop shall be sufficiently tight so that in normal mode of operation during loop maneuver the third gimbal will rotate 180° when the third gimbal axis passes through the vertical in order to keep proper sensing of the servo drive.

3.5.3.1.4 Erection. The erection mechanism shall incorporate separate system for pitch and bank. Erection shall be accomplished by electrolytic switches an torque motors on the gimbals. When displaced from vertical in one axis, the gyro shall not deviate in the other axis by more than 1.0° while it is erectin The erection system shall provide both initial and final erection and shall erect the gyro from any attitude to the vertical as determined by the vertical sensing electrolytic switches. The pitch and roll erection torquer leads shall be brought out of the case so that precession voltage can be applied extern all to precess the gyro for plane rate and earth rate correction.

3.5.3.1.4.1 Initial erection. The vertical gyro shall have fast initial erection so that it will assume a position within 1° of vertical in pitch and bank within 1-1/2 minutes or less after power is applied. The rotor shall be at a standstill and the spin axis may be in any possible attitude at the time power is applied.

3.5.3.1.4.2 Final erection. The vertical gyro shall align its spin axis with the direction of gravity within ±0.25° within 2 minutes or less after power is applied.

3.5.3.1.4.3 Normal erection rate. The nominal rate of normal erection about the roll and pitch axes shall be 1.3° per minute and shall fall between the limits plotted on figure 1 throughout all environments and roll attitudes of the control case. Pitch erection sensing shall be maintained when displacing the gyro in roll ±45° or less.

3.5.3.1.4.4 Fast erection. Means shall be provided to accomplish fast erecti of the gyro to the apparent vertical of both axes of the gyro simultaneously. Fast erection shall be obtained by externally closing the circuit between terminals J and L of table I. The fast erection rate shall be not less than 15° per minute in either axis.

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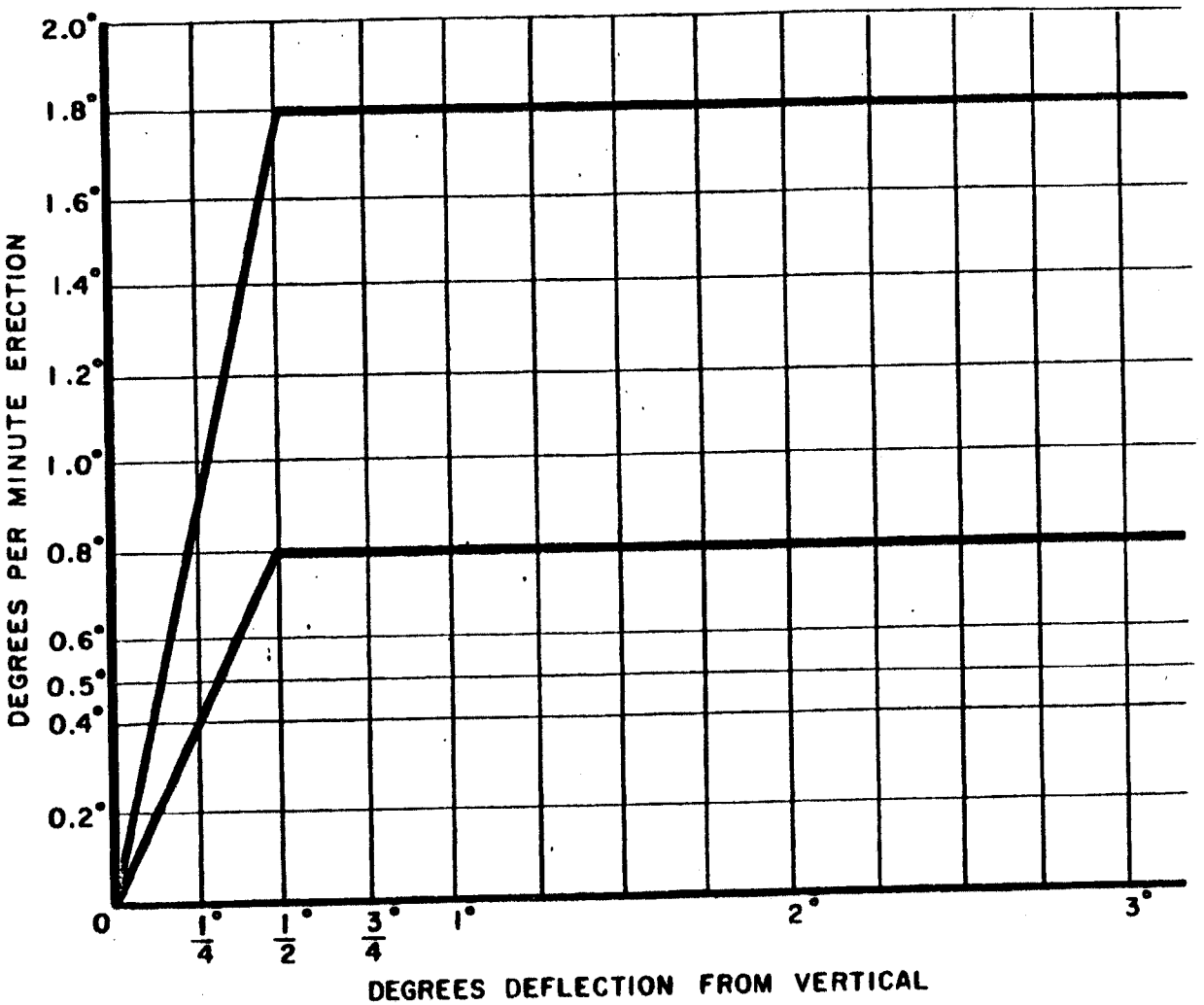


FIGURE 1. Erection Rate As a Function of Deflection

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3.5.3.1.4.5 Roll and pitch pendulum erection cutout. The design shall be such that when pins R and S of table I are connected externally, bank erection will be in effect and when pins H and S of table I are connected externally, pitch erection will be in effect. When all three terminals are commoned externally, both bank and pitch erection shall be functioning. When either pin R or H, or both, are disconnected externally from pin S, bank or pitch or both erection circuits shall be cut out. Pendulum erection cutout shall not affect the torquing of the gyro resulting from input of plane and earth-rate correction.

3.5.3.1.4.6 Fast precession. For test purposes, a means shall be provided to apply and remove, from an external source, a fast precession rate in either direction to the roll axis of the vertical gyro after the gyro is up to speed. This precession rate shall be not less than 15° per minute.

3.5.3.1.5 Vertical gyro synchro outputs. One transmitting synchro shall be mounted on each vertical gyro axis of freedom, including inner roll. These synchros shall provide displacement signals of gimbal movement about their respective axes at a 1-to-1 ratio. The synchros on the vertical gyro axes of freedom shall be excited internally with the voltages across pins D and A of the control electrical connector shown in table I. The stator leads of the synchros shall be brought out through the output connector terminals in accordance with table II. The pitch and roll synchros may be connected to othe control circuitry provided up to 12 high-impedance control transformers, Bendix type AY500-5, or equivalent, with 10K ohm loads can be operated externally from each.

3.5.3.1.5.1 Pitch and outer roll synchro transmitter characteristics

3.5.3.1.5.1.1 Room temperature operation. At room temperature, the pitch and roll transmitters shall have a no-load line-to-line output voltage of 11.8 ±0.25V and no-load phase shift of 4.0° ±1.5°. When loaded with five control transformers (CT'S), Bendix type AY500-5, or equivalent, and with CT rotors open-circuited, the electrical characteristics shall not vary by mor than the following:

a. The voltage drop from the nominal unloaded line-to-line output shall not exceed 0.6V

b. The change in the phase shift angle through the transmitter from the nominal unloaded phase shift angle shall not exceed 3.5°.

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3.5.3.1.5.1.2 Low temperature operation. The displacement gyroscope shall be placed in a chamber at -54° ±2°C for a period of 4 hours. While at this temperature, the transmitters shall meet the following requirements:

a. The no-load output voltage line-to-line shall be no greater than 12.5V and no-load phase shift shall not exceed 5.0°

b. For the five-CT-load condition (rotors open-circuited), the voltage drop from the nominal no-load output shall not exceed 0.9V and the phase shift change shall not exceed 1.5°.

3.5.3.1.5.1.3 High temperature operation. The displacement gyroscope shall be placed in a chamber at a temperature of 71° ±2°C for a period of 4 hours. While at this temperature, the transmitters shall meet the following requirements:

a. The no-load output line-to-line shall be no less than 11.2V. The no-load phase shift shall not exceed 8.1°

b. For the five CT load condition (rotor open-circuited), the voltage drop from the nominal no-load output shall not exceed 0.9V and the phase shift change shall not exceed 3.5°.

3.5.3.1.5.2 Synchro zeroing and sensing. The transmitting synchros on the vertical gyro axes of freedom shall be zeroed, the lead pins shown in tables I and II, connected, and the following procedure used for identification of the pins:

3.5.3.1.5.2.1 Inner roll synchro. Pin A of table I shall be connected to pin A of table II. The voltage across pins A and C of table II and the voltage across pins D of table I and B of table II shall be measured with the gimbals at zero displacement. The voltage across pins A and C of table II shall be at the minimum and the voltage across pins D of table I and B of table II shall be greater than the voltage applied to pins D and A of table I. The voltage across pins B and A and across B and C of table II shall be measured as the first gimbal (inner roll) is held and the reference assembly with the outer roll gimbal is rotated in right roll. The voltage across pins B and C of table II shall increase before decreasing and the voltage across pins B and A of table II shall decrease before increasing.

3.5.3.1.5.2.2 Internally excited pitch synchro. Pin D of table II shall be connected to pin A of table I. The voltage across pins E and F of table II and across pins D of table I and E or F of table II shall be measured with the gimbals at zero displacement. The voltage across pins E and F of table II shall be at the minimum and the voltage across pins D of table I and E or F of

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table II shall be less than the voltage applied to pins D and A of table I. The voltage across pins D and F and across D and E of table II shall be measured as the second gimbal (pitch) is held and the reference assembly with the outer roll gimbal is rotated in climb. The voltage across pins F and D of table II shall increase before decreasing and the voltage across pins E and D of table II shall decrease before increasing.

3.5.3.1.5.2.3 Internally excited outer roll synchro. Pin G of table II shall be connected to pin A of table I. The voltage across pins G and J of table II and across D of table I and H of table II shall be measured with the gimbals at zero displacement. The voltage across pins G and J of table II shall be at the minimum and the voltage across pins D of table I and H of table II shall be greater than the voltage applied to pins D and A of table I. The voltage across pins H and J of table II and across H and G of table II shall be measured as the third gimbal (outer roll) is held and the reference assembly is rotated in right roll. The voltage across pins H and G of table I shall increase before decreasing and the voltage across pins H and J of table II shall decrease before increasing.

3.5.3.1.6 Potentiometer outputs. The displacement gyroscope shall incorporate a single linear potentiometer output on the pitch axis and four sine-cosine potentiometer outputs on the roll axis. The potentiometer assemblies shall meet the requirements of MIL-E-5400 except as specified herein.

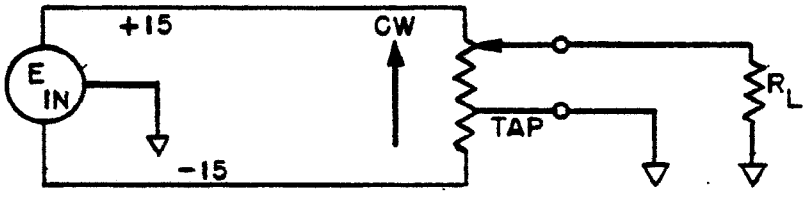
3.5.3.1.6.1 Pitch linear potentiometer output. A single linear potentiometer output shall be provided on the pitch axis which shall incorporate the following characteristics:

a. Terminal resistance: 4.0K ohms ±10 percent

b. Functional angle: 330° ±3°

c. Independent linearity: 0.2 percent

d. Circuit utilization:



(1) R_L: 100K ohms minimum

(2) R_L Connected between wiper and tap (excitation return)

(3) E_{in}: ±15V dc (30V dc total)

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(4) Maximum shaft speed: 60 rpm

(5) Minimum shaft speed: 0 rpm

(6) Servo positioned

e. Tapping requirements:

(1) Type and use: Zero width (1 tap)

(2) Location: 50 percent of excitation, ±0.2 percent

(3) Connected to: Excitation return

f. Power dissipation rating:

(1) 2 watts up to 85°C

(2) Derate to zero at 160°C.

3.5.3.1.6.1.1 Voltage reference excitation. The reference voltage potentiometer excitation shall be +15 and -15V dc as specified in 3.5.3.1.6.1 d(3) and shall be provided from the radar system specified in 3.2.1.2.1.

3.5.3.1.6.1.2 Potentiometer phasing and mechanization. Center tap and function arm shall be presented as the output. The center tap shall be at the 50 percent point ±0.2 percent; 0-V output shall represent 0° in pitch; and positive voltage shall indicate positive pitch (nose up).

3.5.3.1.6.1.3 Potentiometer output accuracy. When the linear potentiometer output is connected to a 200K-ohm load, the linearity of the output shall be 0.20 percent of full scale.

3.5.3.1.6.2 Roll sine-cosine potentiometer outputs. Four sine-cosine potentiometer outputs shall be provided on the roll axis and shall have the following characteristics:

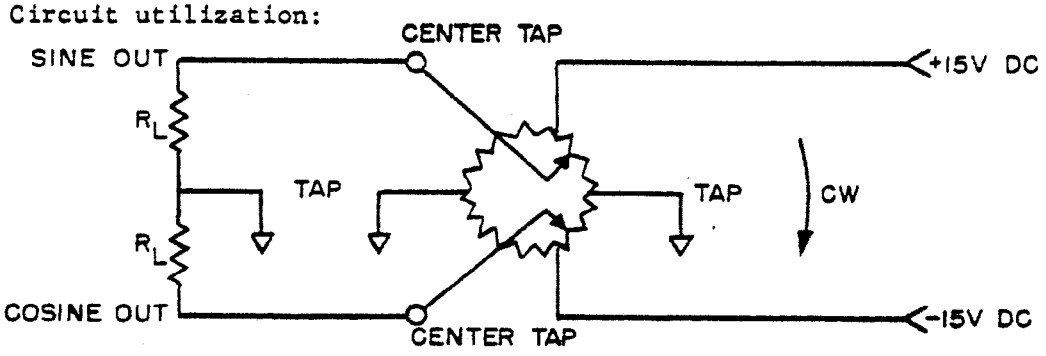
a. Terminal resistance: 4.0K ohms ±10 percent

b. Function angle: 360°

c. Conformity: 0.4 percent

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d. Circuit utilization:



(1) R_L: 100K ohms minimum

(2) R_L: connected between wiper and tap (excitation return)

(3) E_{in}: ±15V dc (30V dc total)

(4) Maximum shaft speed: 60 rpm

(5) Minimum shaft speed: 0 rpm

(6) Servo positioned

e. Tapping requirements:

(1) Type and use: Zero width (2 taps)

(2) Location: 0° and 180° W.R.T. sine slider, ±1.5°

(3) Connected to: Excitation return

f. Power dissipation rating:

(1) 2 watts to 85°C

(2) Derate to zero at 160°C.

3.5.3.1.6.2.1 Voltage reference excitation. The reference voltage potentiometer excitation shall be +15 and -15V dc as specified in 3.5.3.1.6.2d and shall be provided externally from the radar system specified in 3.2.1.2.1.

3.5.3.1.6.2.2 Potentiometer phasing and mechanization. The tapping location shall be 0° and 180° with respect to the sine slider to within ±1.5°. Sine outputs shall be 0V dc and cosine outputs shall be positive when roll is 0°. Sine outputs shall be positive for positive roll (aircraft right wing down). The above condition will exist with +15 and -15V d-c reference excitation. During operation, excitation will not exceed +15 and -15V dc.

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3.5.3.1.6.2.3 Gyroscope potentiometer output accuracy. When the sine and cosine outputs are connected to a 100K-ohm load each, the accuracy shall be 0.42 percent of full scale.

3.5.3.1.7 Vertical gyro drift. Unless otherwise specified, the vertical gyro assembly shall be so designed that the maximum free drift, exclusive of the earth's rotation, will be no greater than 12° per hour with the gyro spin axis vertical and with the spin axis tilted 20° off vertical.

3.5.3.1.8 Power-off freedom. When power is removed momentarily, the vertical gyro shall still have freedom about its first and second gimbal axes and shall not be displaced from the vertical during bank angles of ±60° or pitch angles through ±80°.

3.5.3.2 Directional gyro. The directional gyro shall have sufficient angular momentum and be of such design as to provide the performance specified herein when operated throughout the temperature, vibration, and acceleration ranges specified.

3.5.3.2.1 Gimbal system. The directional gyro shall be suspended in a gimbal system of which the outer gimbal is the third (servoed) gimbal of the vertical gyro. The directional gyro shall have three other gimbals suspended from this outer gimbal.

3.5.3.2.1.1 Inner directional gyro gimbal. The first (inner) directional gyro gimbal (gyro wheel housing) shall be stabilized by the inertia of the gyro. Its pivot axis shall be the first directional gyro axis of freedom, shall be perpendicular to the spin axis of the gyro, and normally shall lie in the horizontal plane. The slaving torquer shall be mounted on this axis. The gimbal shall have a minimum of ±60° freedom about its axis.

3.5.3.2.1.2 Directional gyro azimuth gimbal. The second directional gyro gimbal shall be stabilized by the inertia of the gyro. Its pivot axis shall be the second directional gyro axis of freedom and shall be normally aligned with the vertical. The azimuth output synchros and the leveling torquer shall be mounted on this axis. The gimbal shall be capable of continuous rotation about its axis.

3.5.3.2.1.3 Directional gyro pitch gimbal. The third directional gyro (pitch) gimbal shall be servo driven so that it follows the vertical gyro second (pitch) gimbal at a maximum rate of not less than 200° per second. Its pivot axis shall be the third directional gyro axis of freedom and shall be parallel to the vertical gyro second axis (pitch) of freedom. The gimbal shall be capable of continuous rotation about its axis.

3.5.3.2.2 Directional gyro leveling. The inner gimbal shall be slaved by means of the leveling torquer controlled by a suitable sensor so designed that the spin axis of the directional gyro will be perpendicular to the azimuth axis. This leveling rate shall have an initial rate that will level the gyro within 1-1/2 minutes after power is applied at a normal leveling rate of 1° to 3° per minute.

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3.5.3.2.3 Directional gyro slaving. The directional gyro shall be torqued about its azimuth axis to an external azimuth signal by means of the azimuth torquer. The normal slaving rate shall be 1° to 2° per minute.

3.5.3.2.4 Azimuth transmitting synchros. Two synchros shall be attached to the azimuth axis of the gyro to transmit signals proportional at a 1-to-1 ratio to the position of the gyro. The synchros shall be of the 3-phase wye-connected stator and salient-pole single-phase rotor types.

3.5.3.2.4.1 Azimuth transmitter no. 1. The azimuth transmitter synchro shall be capable of operating up to 12 high impedance CT's, Bendix type AY500-5, or equal, when the CT's are loaded with 10,000-ohm loads. The stator leads of the synchro shall be brought out to the pins of the connector in accordance with table II and shall be separated from other circuitry. The synchro shall be connected internally to neutral and shall be excited by the voltage applied to pin D as shown in table I. The transmitter shall have an electrical accuracy of $\pm$8 minutes or less.

3.5.3.2.4.2 Azimuth transmitter no. 2. The second azimuth transmitter shall operate from a rotor excitation of 115V 400 Hz. The maximum stator output voltage, line-to-line, shall be 11.8V. The transmitter shall have an electrical accuracy of $\pm$15 minutes or less. The rotor and stator leads shall be brought out to separate pins of the electrical connector as shown in tables I and II and shall be isolated from other circuitry.

3.5.3.2.4.3 Synchro zeroing and sensing. The transmitting synchros on the directional gyro axes of freedom shall be zeroed, the lead pins shown in tables I and II, connected, and the following procedure used for identification of the pins:

3.5.3.2.4.3.1 Internally excited azimuth transmitter. Pin K of table II shall be connected to pin A of table I. The voltage across L and M of table II and the voltage across terminals A of table I and L or M of table II shall be measured with the gimbals at zero displacement, the spin axis of the directional gyro aligned with the fore-aft axis of the reference assembly, and with the directional gyro spinning clockwise (cw) when viewed from the aft end. The voltage across pins L and M of table II shall be at the minimum and the voltage across pins D of table I and L or M of table II shall be greater than the voltage applied to pins D and A of table I. The voltage across pins L and K and across M and K of table II shall be measured as the directional gyro second (azimuth) gimbal is held and the reference assembly is rotated about its axis in right turn. The voltage across pins L and K of table II shall increase before decreasing and the voltage across pins M and K of table II shall decrease before increasing.

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3.5.3.2.4.3.2 Externally excited autopilot transmitter. Pins N of table II and F of table I shall be connected to the power lead to pin A of table I and pin G of table I shall be connected to the power lead to pin D of table I. The voltage across pins P and R of table II and the voltage across pins G of table I and P or R of table II shall be measured with the gimbals at zero displacement. The spin axis of the directional gyro shall be aligned with the fore-aft axis of the reference assembly with the directional gyro spinning cw when viewed from the aft end. The voltage across pins P and R of table II shall be at the minimum and the voltage across pin D of table I and pin P or R of table I shall be greater than the voltage applied to pins A and D of table I. The voltage across pins R and N of table II and across P and N of table II shall be measured as the directional gyro second (azimuth) gimbal is held and the reference assembly is rotated about its axis in right turn. The voltage across pins R and N of table II shall increase before decreasing and the voltage across pins P and N of table II shall decrease before increasing.

3.5.3.2.5 Balance. The directional gyro shall be so balanced that the apparent drift due to the earth's rotation will be equal to the product of the sine of the latitude of operation and 15° per hour in a counterclockwise (ccw) direction in the Northern Hemisphere and in a cw direction in the Southern Hemisphere.

3.5.3.2.6 Random drift. The random drift of the directional gyro shall not exceed $\pm 0.5^{\circ}$ per hour (rms) from the computed apparent drift for the latitude of operation.

3.5.3.2.7 Maximum drift. The maximum drift from the computed drift shall not exceed $\pm 1^{\circ}$ per hour.

3.5.3.3 Dimensions. The dimensions of the displacement gyroscope shall be as shown on figure 2.

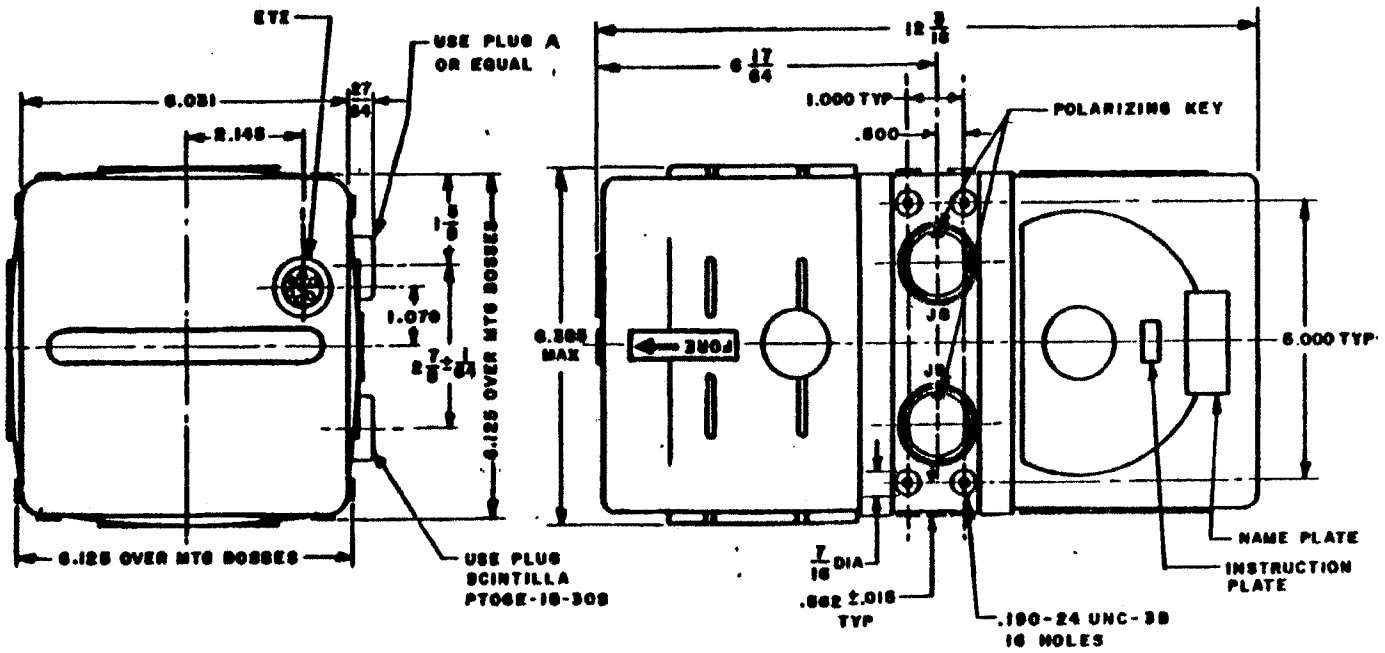
3.5.3.4 Weight. The weight of the displacement gyroscope shall not exceed 31.5 pounds.

3.5.3.5 Electrical connectors. The electrical connectors shall be in accordance with tables I and II and figure 3.

3.5.3.6 Mounting. The mounting design shall be such as to enable the displacement gyroscope to meet the vibration failure, acceleration, and shock tests.

3.5.3.7 Reference amplifiers. The reference amplifiers shall be contained integrally within the reference package and shall fulfill all necessary servo and power requirements for proper reference operations. The reference package shall contain all amplifiers and electrical components for the servoed gimbals, the pitch and azimuth servos, for related equipment, and for the starting and erection controls.

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PLUG	
A	B
PT06E-18-32SW	PT06E-18-30SW

FIGURE 2. Displacement C oscope Dimensions

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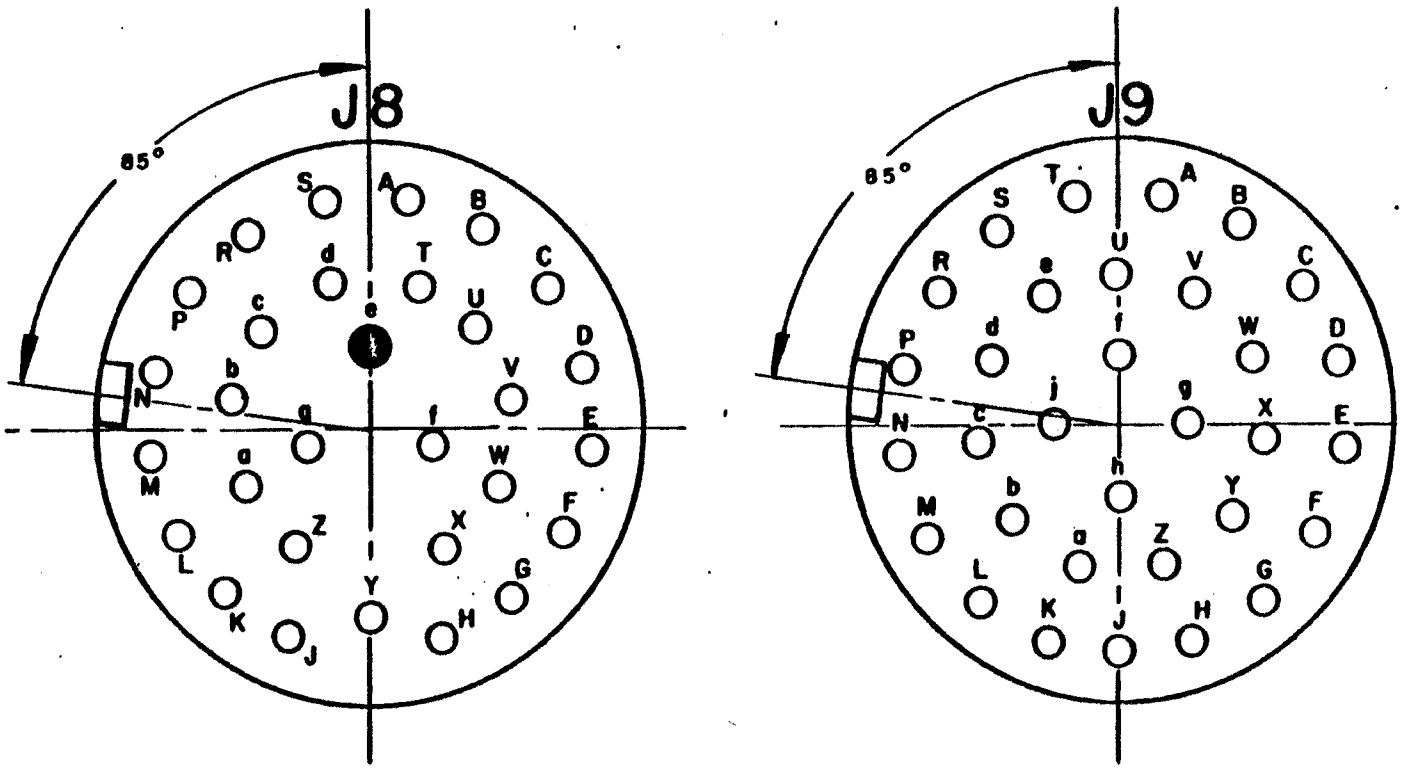


FIGURE 3. Pin Identification, Displacement Gyroscope

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3.5.3.7.1 Amplifier channels. The amplifiers shall not incorporate electronic tubes. Parallel circuits may be used where practicable so that partial failure will not result in total loss of operation.

3.5.3.7.2 Starting and erection control. The starting and erection control shall be designed to perform the functions of starting sequence control, fast erection control, power failure warning, roll erection cutout, and pitch erection cutout.

3.5.3.7.2.1 Starting sequence control. The starting sequence control shall properly align the gimbals of the reference assembly and erect and align the gyro as required in 3.5.3.1.4.1, 3.5.3.1.4.2, and 3.5.3.2.2 when the servoed gimbals are in any initial position and when the system power has been disconnected long enough to cause recycling even though the gyros may still be coasting. Proper alignment shall also occur when the servoed shafts and gimbals of the systems are rotated to any new position after power is disconnected.

3.5.3.7.2.2 Power failure warning. A 3-phase power-warning circuit shall be incorporated in the system which shall operate a set of normally open contacts. Leads from this set of contacts shall be brought out of the case to pins W and X of table I, and shall be separated from other circuitry. When proper power is applied to the control, the contacts shall close the circuit between pins W and X after the initial erection cycle is completed. If at any time any p' `e of the 3-phase power to the control is broken, the contacts shall open ar. rea the circuit between pins W and X.

3.5.3.8 Third gimbal operation. The following provisions shall be made for operation of the third gimbal:

3.5.3.8.1 Pitch drive. The rotor of a resolver synchro shall be connected to the pitch gimbal servo. The pitch servo shall be designed to follow the pitch input 1-to-1 at a rate of not less than 200° per second.

3.5.3.9 Time totalizing meter (TTM). The gyroscopes shall incorporate a digitalized 9,999-hour TTM in accordance with MS17322-8, or equivalent, to record the operating time of the units. The indicator shall be so located that it will not be necessary to disassemble the gyroscope case in order to read the elapsed time.

3.5.4 Electronic control amplifier

3.5.4.1 Design and construction. The ASK-8A/A24G-5A electronic control amplifier shall be designed to supply aircraft heading to autopilots, flight director systems, and other aircraft equipment requiring heading information. The amplifier shall contain the proper circuitry to receive, amplify, modify, and transmit heading information received from a magnetic and an inertial sour

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3.5.4.1.1 Compatibility. The electronic control amplifier shall be electrically and physically interchangeable with the J-4 tube version and the ASK-8/A24G-5 tube version. It shall be compatible with the J-4 and AJN-3 compass systems, the A/A24G-5 control assembly conforming to MIL-C-27205, and with the DT-309/AJN detector conforming to MIL-D-26503.

3.5.4.1.2 Subassemblies and component parts. The electronic control amplifier shall consist of the following plug-in subassemblies:

a. Power Supply - Shall operate with power source frequency of 300 to 500 Hz

b. Mechanism - The mechanism unit shall contain the following rotating components:

(1) Compass control transformer

(2) Azimuth control transformer

(3) Azimuth motor-generator with gear train

(4) Slew motor-generator with gear train

(5) Two data transmitters

c. Transistorized slaving amplifier

d. Transistorized azimuth amplifier

e. Relay module for automatic synchronization, slaving and leveling cutout of associated gyro

f. Wiring harness that electrically interconnects subassemblies and electrical receptacles.

3.5.4.1.2.1 Hermetic seal. The mechanism shall be hermetically sealed to make it impervious to environmental conditions.

3.5.4.1.3 Power requirements. The amplifier shall operate from 115V single-phase 400-Hz a-c and 27.5V d-c power.

3.5.4.1.4 Electrical transients. The electronic control amplifier shall be protected against high-voltage spikes as high as 200V dc for microsecond duration and also against transients of longer duration as specified in MIL-STD-704.

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3.5.4.2 Data transmitters. Two data transmitters with rotors that are positioned by the azimuth servo channel shall be provided for the electrical transmission of heading reference to auxiliary equipment. All connections to rotors and stators shall be brought out to separate pins in the electrical connector.

3.5.4.2.1 Indicator transmitter. A 4-power transmitter with an electrical accuracy of 10 minutes or less shall be provided for operating remote indicators. The rotor excitation shall be 115V 400 Hz and the maximum output voltage, line-to-line, shall be 18.2V, open circuit.

3.5.4.2.2 Precise data transmitter. A transmitter having an electrical accuracy of 7 minutes or less shall be provided for the operation of precise control synchros, Bendix type AY200 series, or equal. The transmitter shall be capable of operating two ID-249/ARN, ID-351A/ARN, or ID-387/ARN, two ID-250A/ARN and two ID-251/ARN course indicators simultaneously. It shall have the following characteristics:

a. Input voltage: 26V 400 Hz

b. Input impedance: 25 +j115 ohms

c. Output voltage (maximum line-to-line): 11.8V.

3.5.4.3 Amplifier and servo channels

3.5.4.3.1 Amplifier channels. The amplifier channels shall be suitable for continuous operation for extended periods of time under all possible service conditions and shall utilize transistors in accordance with MIL-E-5400.

3.5.4.3.2 Servo channels. The servo channels shall so control the servo motors that the response of the compass will be in accordance with the conditions specified herein.

3.5.4.3.3 Azimuth servo channel. The electronic control amplifier shall contain a fast servo system with a followup rate of at least 7 rpm to maintain synchronization with gyro movement. For velocities of 5 rpm or less, the error shall be not greater than 0.2° per rpm. The undamped natural frequency shall be 6 Hz with damping approximately 60 percent of critical.

3.5.4.3.4 Slaving channel. The slaving channel shall incorporate a control transformer which receives a signal from a DT-309/AJN detector conforming to MIL-D-26503, or electrical equivalent, and the rotor of which is positioned

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by the azimuth servo channel. The output of the slaving channel amplifier shall slave the displacement assembly directional gyroscope at the rates specified in 3.5.3.2.3.

3.5.4.4 Slaving and leveling cutout. Provisions shall be made for leveling cutout and for deslaving the gyro. This shall be accomplished by applying 28V dc to two of the terminals of the electrical power connector. Leveling cutout shall operate under all modes of operation. Slaving cutout shall be effective when the compass is in the magnetic mode.

3.5.4.5 Automatic magnetic synchronization. The azimuth servo channel shall be synchronized automatically with the DT-309/AJN detector, or electrical equivalent, upon the initial application of power to the compass when the control panel function selector switch is in the MAG position. The compass shall also be synchronized automatically when it is switched from the DG to the MAG mode of operation. Automatic synchronization shall be accomplished at a rate of approximately 7 rpm.

3.5.4.6 TTM. The amplifier shall incorporate a TTM as specified in 3.5.3.9.

3.5.4.7 Electrical connector function. The electrical connectors shall function as indicated in table III and shall be in accordance with 3.10 and figures 4 and 5.

3.5.4.8 Test terminals. External test terminals shall be provided as indicated in table IV to simplify maintenance and trouble shooting.

3.5.4.9 Case. The outline dimensions of the electronic control amplifier case shall conform to figure 4.

3.5.4.10 Shock mount. The amplifier shall be provided with a rack-type vibration mount in accordance with figure 4.

3.5.4.10.1 Sway. When subjected to vibration as specified herein, the amplifier shall not sway more than the maximum shown on figure 4.

3.5.4.11 Weight. The weight of the complete electronic control amplifier, including mount, shall not exceed 12 pounds.

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TABLE III. ELECTRONIC CONTROL AMPLIFIER CONNECTORS

Connector	Cannon No.	Pin	Function
J2001	K02-16-10SN, or equal	A	To magnetic
		B	detector (DT-173/AJN) or (DT-309/AJN)
		C	
		D	Magnetic detector excitation
		E	
		F, H, J, K, L	Spares
J2002	K02-21-30SN, or equal	A	To gyro
		B	Azimuth output
		C	
		D	Indicator transmitter
		E	Excitation
		F	
		H	To indicator
		S	Repeater
		J	Precise transmitter
		K	Excitation
		L	
		M	To precise repeater
		N	
		P	
		R	Slewing signal
		T	
		f	Gyro, turn erection cutout
		U	
		W	To slew switch
		V	
		d	28V dc to gyro brake
		X	Slaving turn cutout
		Y	Function selector sw
		Z	Latitude correction voltage
		a	Function selector sw
		b	
		c	Spare
		g	Function selector sw

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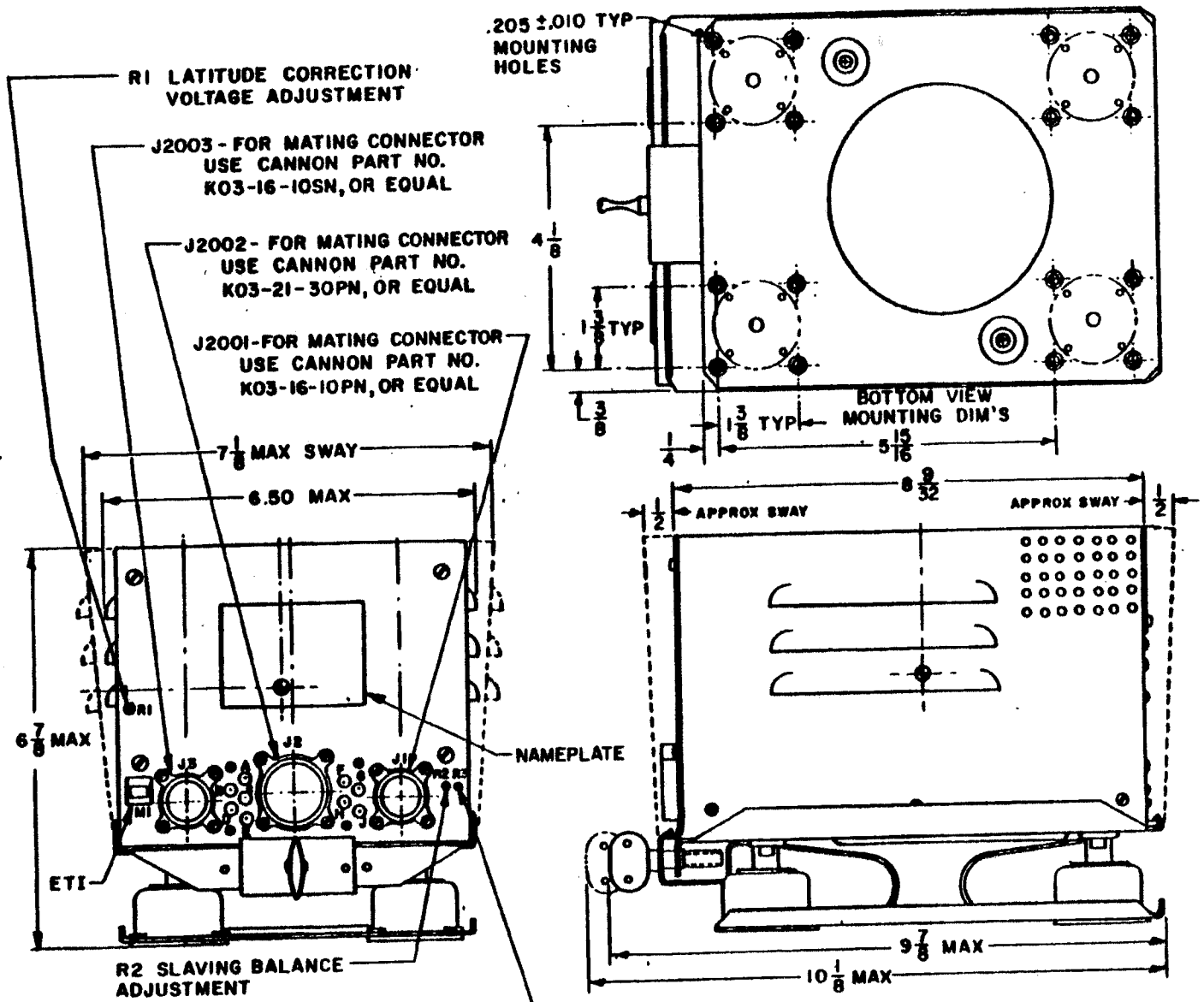
TABLE III. ELECTRONIC CONTROL AMPLIFIER CONNECTORS (CONT'D)

Connector	Cannon No.	Pin	Function
J2002	K02-21-30SN, or equal	h i	Gyro azimuth transmitter excitation
J2003	K02-16-10PN, or equal	A B	115V, 400 Hz excitation
		C D	28V d-c excitation
		E F	Turn cutout circuit (apply 28V dc for turn cutout)
		H	Spare
		J K L	Autopilot cutout

TABLE IV. TEST TERMINALS

Jack	Color	Function
A B	Green (high) Black (low)	Connected to the first voltage amplifier stage in the azimuth amplifier
B C	Blue Blue	Connected to the output of the azimuth amplifier
D E	Gray (high) Black (low)	Connected to the regulated latitude correction voltage output of the power supply
F E	Brown (high) Black (low)	Connected to the pre-amplifier output of the slaving amplifier
G H	Yellow Yellow	Connected to the demodulator output of the slaving amplifier
J E	Red (high) Black (low)	Connected to the power supply inverter output

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FIGURE 4. Electronic Control Amplifier

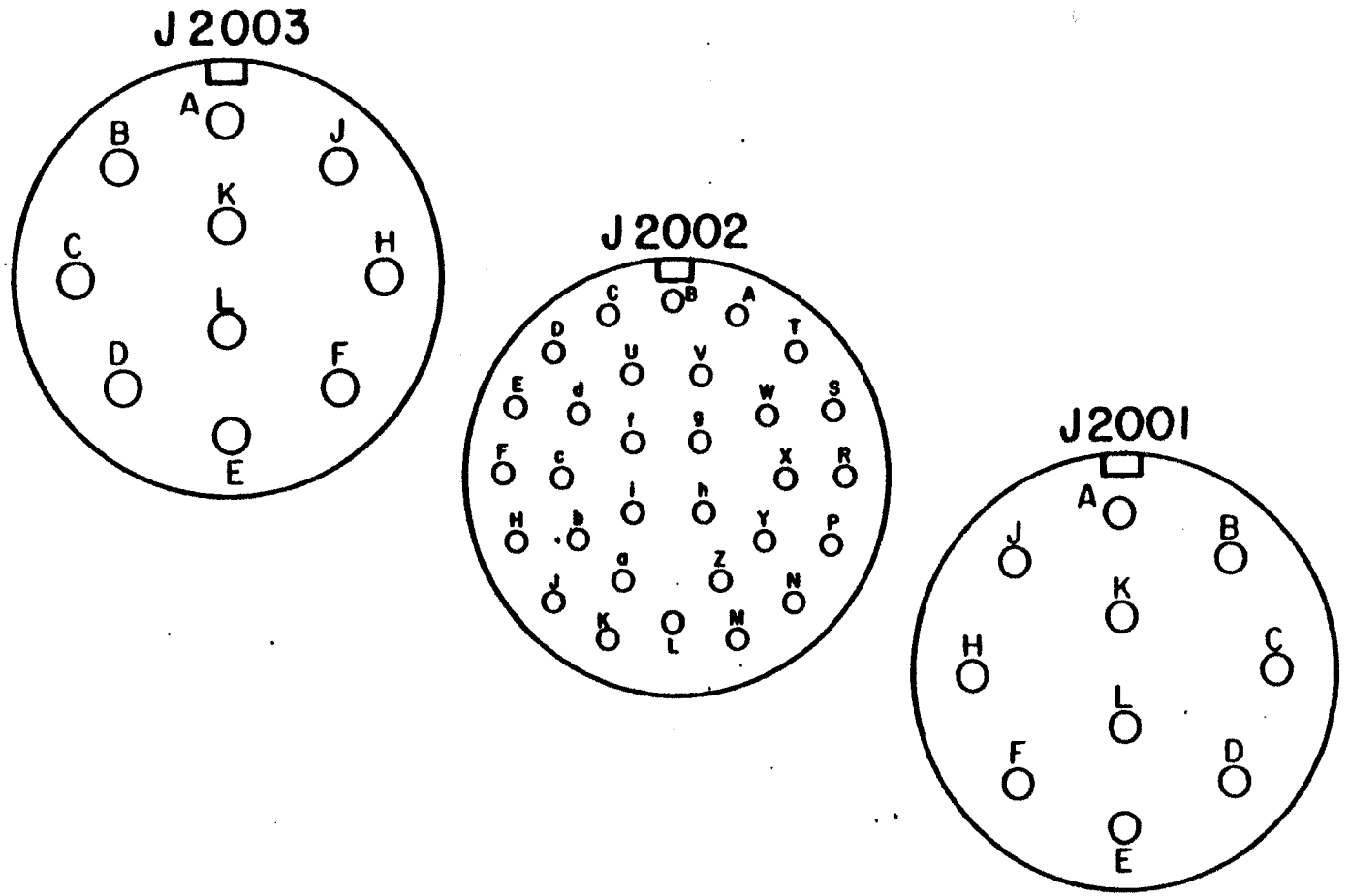


FIGURE 5. Pin Identification, Electronic Control Amplifier

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3.5.5 Function selector switch. The design of the control assembly shall be such that the function selection of MAG or DG mode may be made and the synchronization specified in 3.5.4.5 may be accomplished by means of a transposition switch.

3.5.5.1 The design of the assembly shall be such that it will operate with the control panel of the AN/AJN-3A compass system conforming to MIL-C-25773.

3.5.6 Synthetic jewel bearings. When synthetic jewel bearings are used, they shall conform to MIL-B-27497.

3.6 Performance. The control assembly shall be capable of satisfactory operation when subjected to the following conditions to the extent specified herein:

a. Temperature: Operating temperatures ranging from -54° to +71°C and storage temperature ranging from -65° to +95°C

b. Humidity: Relative humidity up to 100 percent

c. Altitude: Pressures ranging from 30 inches Hg down to 0.809 inch Hg (approximately 80,000-foot altitude)

d. Salt fog: Salt fog for a period of 48 hours

e. Vibration: 5 to 10 Hz, 0.4 inch double amplitude; 10 to 175 Hz, 2g; 175 to 325 Hz, 0.001 inch double amplitude; 325 to 525 Hz, 5g

f. Fungus: Exposure to fungus enzymes

g. Dust: Exposure to blowing fine sand and dust particles

h. Rain: Exposure to simulated rainfall

i. Acceleration: Acceleration forces of 20g for 1 minute

j. Shock: Impact shocks of 20g (operating) and 40g (crash safety) for the gyroscope and 15g (operating) and 30g (crash safety) for the amplifier

k. Thermal shock: Cyclic immersion in temperatures of 71° ±2°C and -54° ±2°C

l. Explosion: Explosive atmosphere.

3.6.1 Service life. The control assembly shall have a service life of 1,500 hours.

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3.6.2 Electromagnetic interference and susceptibility. Electromagnetic interference and susceptability requirements shall be in accordance with MIL-STD-461 for class A1 equipment.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-100 shall govern the manufacturer's part numbers and changes thereto.

3.8 Electronic components. Electronic components shall be in accordance with MIL-E-5400.

3.9 Electrical power. Unless otherwise specified, the power required by the system shall be 4-wire, 200/115V 3 phase, 400 Hz (nominal) in accordance with MIL-STD-704 and as specified herein.

3.9.1 Electrical power variation. The system shall operate properly for any a-c voltage from line-to-neutral between 103 and 127V for any frequency between 320 and 480 Hz, or any combination thereof.

3.9.2 Phase rotation. Phase rotation of the 400-Hz power required by the system shall be A-B-C, with neutral as ground.

3.10 Electrical connectors. Unless otherwise specified, the electrical connectors shall be in accordance with MIL-C-5015.

3.10.1 Test terminals. Adequate test terminals shall be provided on each component to externally test the item.

3.11 Hermetic sealing. The following requirements shall apply except in those extreme cases where the filling medium or the pressure, or both materially affect the performance of the instrument. Exceptions shall be based upon sound engineering facts and shall be presented to the procuring activity for approval.

3.11.1 Case. The dimensions of the gyroscope case shall be as shown on figure 2. The case shall be hermetically sealed and of such design that the internal mechanism may be removed, replaced, and the case resealed without the use of special tools and fixtures unless they are approved by the procuring activity. Hermetic sealing shall be so accomplished that the seal will not be dependent upon materials that would be affected by the action of any atmosphere to which the gyroscope may be subjected. The gyroscope case shall be gray, color No. 36231 of FED-STD-595.

3.11.1.1 Filling medium. The filling medium shall be 95 percent helium and the remainder carbon dioxide, shall be of at least 98 percent purity, free of dust

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particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65°C) at the filling pressure. The absolute pressure of the filling medium in the case shall be 12 to 14.7 psi.

3.12 Soldering. Soldering shall be accomplished in accordance with MIL-STD-454, requirement 5.

3.13 Screw threads. Unless otherwise specified, the threads of all machine screws shall conform to MIL-S-7742.

3.14 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.15 Workmanship. The control assembly, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particula attention shall be given to neatness and thoroughness of soldering, brazing, painting, riveting, machine-screw assemblies, and freedom of parts from burrs and sharp edges.

3.15.1 Dimensions and tolerances. Dimensions and tolerances not specified shall be as close as is consistent with the best shop practices. Where dimensions and tolerances may affect the interchangeability, operation, or performance of the system, they shall be held or limited accordingly.

3.15.2 Screw assemblies. Assembly screws and bolts shall be tight. The d tight means that the screw or bolt cannot be appreciably tightened further without damage or injury to the screw or bolt or threads.

3.15.3 Riveting. Riveting operations shall be performed carefully to insure that the rivets are tight and headed satisfactorily.

3.15.4 Gears. Gear assemblies shall be aligned properly and meshed and shall operate without interference, tight spots, loose spots, or other irregularitie Where required for accuracy adjustment, gear assemblies shall be free from backlash.

3.15.5 Cleaning. The control assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, or other foreign material before and after assembly. Burrs and sharp edges as well as resin flash which might crumb shall also be removed.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of al

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inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of control assemblies (or components) shall be classified as follows:

a. Preproduction tests

b. Quality conformance tests.

4.3 Test conditions

4.3.1 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the test are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately +25°C).

4.3.2 Attitude. Unless otherwise specified, the components shall be tested in their normal operating position.

4.3.3 Properly connected equipment. When the assembly or component is referred to as being connected properly, it shall be understood that the components of the system are connected for normal operation which shall include operation of the servoed gimbal specified in 3.5.3.1.3. The control panel of the AN/AJN-3A compass system shall be used in conducting the tests. The displacement gyroscope shall be hard mounted in all tests. Simulators may be used for components of the system that are not specifically being tested.

4.3.4 Power. Unless otherwise specified, power applied to the system shall be as specified in 3.9.

4.3.5 Measurements. Unless otherwise specified, electronic control amplifier voltage and frequency measurements shall be made in the slaved mode.

4.4 Preproduction testing

4.4.1 Test samples. The test samples shall consist of nine control assemblies or components, as applicable, representative of the production equipment. Three of the control assemblies or components shall be subjected to all the tests except reliability and the other six shall be subjected to the reliability test

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only. The samples shall be identified by the manufacturer's part number and such other information as required by the procuring activity. The contractor shall submit the samples that were tested to the procuring activity.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2).

4.4.3 Preproduction tests. The preproduction tests shall consist of all the tests described under 4.6.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests.

4.5.1 Individual tests

4.5.1.1 Displacement gyroscope. Each displacement gyroscope shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Infant mortality aging

c. Initial erection

d. Final erection

e. Erection rate

f. Erection cutout

g. Manual fast erection

h. Loop test

i. Gimbal followup

j. Roll, pitch, and yaw

k. Vertical gyro drift

l. Power interruption

m. Power failure warning

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n. Azimuth slaving rate and cutout

o. Random wander

p. Latitude correction

q. Balance

r. Power consumption

s. Synchro zeroing and sensing

t. Potentiometer outputs

u. Case leakage.

4.5.1.2 Electronic control amplifier. Each electronic control amplifier shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Infant mortality aging

c. Insulation resistance

d. Continuity

e. Automatic synchronization

f. Excitation voltage

g. Power consumption

h. Flux valve excitation

i. Data transmitter nulls

j. Slaving polarity

k. Data transmitter polarities

l. Slewing polarities

m. Slaving balance

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n. Slaving operational test

o. Autopilot cutout

p. Manual synchronization

q. Slow manual synchronization

r. Test jacks

s. Latitude correction voltage

t. Azimuth servo-amplifier operation

u. Azimuth followup rate

v. Index error

w. E-Z check

x. Turn cutout relay

y. Scale error.

4.5.2 Sampling tests

4.5.2.1 Sampling plan A. One gyroscope and one amplifier shall be selected at random from each 100 or less produced on the contract or order and subjected to the following tests as applicable and as described under 4.6:

a. Individual tests

b. Voltage and frequency variation

c. Low temperature operation

d. High temperature operation

e. Dielectric strength

f. Vibration error

g. D-C voltage variation (electronic control amplifier only)

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4.5.2.2 Sampling plan B. Unless otherwise specified, 3 gyroscopes and 3 amplifiers shall be selected at random from the first 15 items on the contract or order and subjected to the following tests in the sequence specified, as applicable, and as described under 4.6:

a. Sampling plan A tests

b. Electromagnetic interference and susceptibility

c. Temperature - altitude

d. High temperature exposure

e. Low temperature exposure

f. Temperature shock

g. Vibration failure

h. Acceleration

i. Shock

j. Humidity

k. Fungus

l. Rain

m. Salt fog

n. Dust

o. Explosion.

4.5.2.2.1 The following tests may be conducted on empty hermetically sealed cases that have all external parts installed as on a complete unit:

a. Humidity

b. Fungus

c. Rain

d. Salt fog

e. Dust.

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4.5.2.3 Sampling plan C (system compatibility tests). The initial 10, and 1 out of each 25 components manufactured under a specific manufacturer's type number shall be connected as a system and subjected to the following tests as described in 4.6.10.1 through 4.6.10.9 after having satisfactorily met the individual tests. A simulated magnetic detector may be used.

a. Azimuth servo operation

b. Azimuth followup rate

c. Azimuth slaving rate and cutout

d. Polarity

e. Static settling

f. Dynamic accuracy

g. Scale error

h. Synchronization

i. Heading set slew.

4.5.2.3.1 While still connected as a system, and after satisfactorily meeting the tests specified in 4.5.2.3, 2 of the 10 systems shall be subjected to the following tests as described in 4.6.10.10 through 4.6.10.12.

a. Voltage and frequency variation

b. Low temperature operation

c. High temperature operation

4.5.2.4 Sampling plan D. Six displacement gyroscopes and six electronic control amplifiers selected at random from the first 15 items on the contract or order shall be subjected to the reliability test specified in 4.6.12.

4.5.2.5 Rejection and retest. When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined.

4.5.2.5.1 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

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4.5.3 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 Test methods

4.6.1 Displacement gyroscope

4.6.1.1 Examination of product. The displacement gyroscope shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, weight, and outline dimensions. The results shall be recorded.

4.6.1.2 Infant mortality aging. The displacement gyroscope shall be operated for 48 hours while mounted on a Scorsby-type mechanism having a 15° movement at 5 to 7 oscillations per minute. Outputs and inputs shall be loaded to simulate actual use. In case of failure or an out-of-tolerance condition, the failed component(s) shall be replaced, or the out-of-tolerance condition corrected, and the gyroscope then subjected to the applicable individual tests prior to continuing this test. This test need not be repeated except as necessary to complete the 48-hour aging period. If additional failures or out-of-tolerance conditions occur during the aging period, they shall be corrected as specified herein. Upon completion of this test, the gyroscope shall be subjected to the remaining individual tests.

4.6.1.3 Initial erection. The displacement gyroscope shall be connected properly for normal operation without power applied. The displacement reference shall be mounted to a test base and leveled to within 1/10° of the horizontal as determined by the local gravity vertical. Power shall be applied to the system. Within 1-1/2 minutes, the gyros shall be erected to within 1° of their proper relationship to the local gravity vertical, the azimuth circuit shall be synchronized, the erection circuits shall switch to slow erection and continuity shall exist between terminals W and X of table I. After application of power, all operation shall be automatic.

4.6.1.4 Final erection. Within 2 minutes after initial application of power, the vertical gyro shall be erected to within 1/4° of the local gravity vertical which shall be determined by the synchro transmitter outputs. The displacement gyroscope shall be rotated rapidly 180° in azimuth. The indication of pitch and bank shall not change by more than 1/2°. During this test, the displacement gyroscope may be subjected to the motions of a low-angle (not to exceed 2° total movement) roll, pitch, and yaw mechanism operated at a speed of 5 to 7 oscillations per minute or vibrations of 0.004 to 0.006 inch total amplitude at 10 to 12 Hz.

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4.6.1.5 Erection rate. The vertical gyro shall be deflected between 45° and 20° in right bank. The fast erection shall be applied and the time required for the gyro to erect 15° shall be recorded; the fast erection shall be removed and the time required for the gyro to erect 5° on normal erection shall be recorded. The computed fast erection rate shall be not less than 15° per minute and the computed normal erection rate shall be between 0.80° and 1.8° per minute. While erecting, the gyro shall not deviate in the undeflected axis by more than 1°. This test may be conducted with the displacement gyroscope subjected to the motions of a low-angle (2° total movement) roll, pitch, and yaw mechanism. The same test shall be conducted for left bank, dive, and climb and the same tolerances shall apply.

4.6.1.6 Erection cutout. The vertical gyro shall be displaced approximately 15° in right bank and either direction in pitch. The bank erection shall be disconnected externally and the gyro shall continue to erect in pitch but shall cease to erect in bank. Bank erection shall be reconnected and pitch erection disconnected externally. The gyro shall erect in bank but cease to erect in pitch. This test shall be repeated with deflections in the opposite direction.

4.6.1.7 Manual fast erection. With the vertical gyro deflected as specified in 4.6.1.5, terminals L and J of table I shall be connected. The gyro shall erect in each direction at a minimum rate of 15° per minute.

4.6.1.8 Loop test. The displacement gyroscope shall be mounted on a fixture permitting 360° rotation about a horizontal axis so that the pitch axis of the vertical gyro is parallel to the fixture axis.

4.6.1.8.1 Normal operation. The displacement gyroscope shall be connected properly for DG mode, normal operation, power shall be applied, and erection completed. It shall be rotated through one complete revolution (loop) at a rate of 30° per second. The displacement gyroscope shall be leveled immediately and displacement of the gyro noted. The vertical gyro shall be erected to within 1/2° of the local gravity vertical. The directional gyro shall not have deviated more than 1/2°.

4.6.1.9 Gimbal followup. The displacement gyroscope shall be so positioned on the fixture that it may be rotated about its roll axis. The displacement gyroscope shall be connected properly and shall be operating. It shall be tested to and shall meet the servo response and operation requirements of 3.5.3.1.3. The roll gimbal steady-state lag error shall not exceed 15°.

4.6.1.10 Roll, pitch, and yaw. The displacement gyroscope shall be connected for normal operation on the roll, pitch, and yaw mechanism and shall be operated at 5 to 7 oscillations per minute with a total amplitude of 15° in

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each of the 3 axes for a period of 15 minutes. The roll, pitch, and yaw mechanism shall be leveled within $\pm 1/10^{\circ}$ and the indication read immediately. The vertical gyro shall be within $1/2^{\circ}$ of the local gravity vertical as indicated by the output synchro signals.

4.6.1.11 Vertical gyro drift. The vertical gyro drift test shall be conducted as follows:

a. The displacement gyroscope shall be positioned on the roll, pitch, and yaw mechanism so that its fore and aft axis is pointed to the geographic east. The spin axis of the vertical gyroscope shall be aligned with gravity (vertical) and the pitch and roll erection shall be disconnected. The roll, pitch, and yaw mechanism shall be operated at a speed of 5 to 7 oscillations per minute and a total amplitude of 2° for at least 10 minutes. With the proper correction for the earth's rotation, the drift rate in either axis shall not exceed 12° per hour.

b. Step a shall be repeated except the displacement gyroscope shall be pointed to the north heading.

c. Steps a and b shall be repeated except the vertical gyro shall be tilted 20° in an easterly heading and the displacement gyroscope shall be pointed to the north, west, and south headings. The same tolerances and drift rate shall apply.

4.6.1.12 Power interruption. The displacement gyroscope shall be connected properly and after 5 minutes of operation, the power shall be removed and the displacement gyroscope moved through angles of $\pm 60^{\circ}$ in pitch and $\pm 60^{\circ}$ in roll and returned to level. Power shall be applied 40 seconds after removal. The pitch and bank axes shall indicate level within $\pm 1^{\circ}$ and there shall be no recycling of the initial erection. The circuit between pins W and X of table I shall close immediately.

4.6.1.13 Power failure warning. The displacement gyroscope shall be connected properly and power applied. The time required for the internal circuit to close between pins W and X of table I shall not exceed 90 seconds. Each phase of the input power to the system shall be broken individually and the internal circuit between pins W and X shall open. The broken power leg shall be reconnected and the internal circuit between W and X shall close immediately.

4.6.1.14 Azimuth slaving rate and cutout. The displacement gyroscope and the system simulator shall be set into operation in the MAG mode and synchronized with the magnetic detector on north heading. The magnetic detector shall then be turned approximately 30° cw and the displacement through which the gyro will be torqued shall be between 10° and 20° for a 10-minute period with the

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proper correction for the earth's rotation. External deslaving connection shall be made in accordance with 3.5.4.4 and torquing on the directional gyro shall cease; it shall continue when the connections are broken. The function selector switch shall be turned to DG position and the directional gyro shall cease to be torqued. The slaving rate and cutout tests shall be repeated with the detector turned initially approximately 30° in a cw direction. The same tolerances shall apply.

4.6.1.15 Directional gyroscope random wander. The displacement gyroscope shall be connected properly and operating on a 2° total movement roll, pitch, and yaw mechanism which provides 5 to 7 oscillations per minute. The gyros shall be up to speed and erected. Slaving and latitude correction shall be cut out and the mechanism shall be made to oscillate. The extremes of the indications of the azimuth displacement shall be read and recorded for each 1/2-hour period for a total time of not less than 4 hours. The hourly drift rate for each 1/2-hour period shall be determined and shall not differ from the computed drift for the local latitude by more than $\pm 1.0^{\circ}$ per hour. The rms value of the eight readings shall not exceed $\pm 0.5^{\circ}$ per hour when each reading is corrected for the earth's rate at the local latitude.

4.6.1.16 Latitude correction. The displacement gyroscope shall be operated with the latitude correction control set for periods of not less than 1/2 hour each at 0°, 60°N, 90°N, and 90°S. It shall then be tested in accordance with 4.6.1.15. The drift of the gyro for each hourly period shall not differ by more than $\pm 2^{\circ}$ per hour from the difference between the drift computed for t local latitude and the drift correction computed for the latitude of correction control setting.

4.6.1.17 Balance. The displacement gyroscope shall be oriented with fore end in a north heading and leveling, slaving, and latitude correction cutout. The directional gyro pitch gimbal shall be so displaced that the azimuth axis is aligned perpendicular to a plane which is equivalent (parallel) to the tangent of the earth at the equator. The gyro spin axis shall be aligned north-south and perpendicular to the azimuth axis (parallel to earth's rotational axis). From this position, the drift shall not exceed 5.5° for a 1-hour period when subjected to the test specified in 4.6.1.15. This test may be accomplished prior to final sealing of the case.

4.6.1.18 Power consumption. The system shall be connected for normal operation without power applied. The power consumption of the system shall be checked under starting and running conditions. The power consumption shall not exceed 350 va during starting and 225 va during running.

4.6.1.19 Synchro zeroing and sensing. Each transmitter, control transformer, and differential synchro of the system shall be tested and the zeroing and sensing shall be in accordance with the respective requirements of 3.5.3.1.5.2.

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4.6.1.20 Potentiometer outputs

4.6.1.20.1 Pitch linear. The pitch linear potentiometer output shall be tested as follows:

a. Place the displacement gyroscope on a tilt table in such a manner that the gyro may be rotated about the pitch axis. Connect the potentiometer excitation as per table I and connect pin f of table II and pin P of table I to voltage ratio equipment with a 200K-ohm load connected in parallel.

b. Connect the displacement gyroscope pitch synchro output to a precision angle indicator

c. Apply power and allow system to stabilize

d. Tilt the table about the pitch axis until the voltage ratios listed in table V are obtained. The displacement of the gyro about the pitch axis, as read on the precision angle indicator, shall be as specified therein.

TABLE V. PERCENT VOLTAGE RATIO FOR PITCH

Input Voltage Ratio	Output Pitch Synchro				
		Minimum		Maximum	
Nominal	Nominal	Degrees	Minutes	Degrees	Minutes
0	0	-0	54	0	54
3.6364	+60	59	6	60	54
0.9090	+15	14	6	15	54
0.6060	+10	9	6	10	54
0.3030	+5	4	6	5	54
0	0	-0	54	0	54
0.3030	-5	-5	54	-4	6
0.6060	-10	-10	54	-9	6
0.9090	-15	-15	54	-14	6
3.6364	-60	-60	54	-59	6
0	0	-0	54	0	54

4.6.1.20.2 Sine-cosine roll. The sine-cosine roll potentiometer outputs shall be tested as follows:

a. Place the displacement gyroscope on a tilt table in such a manner that it will rotate the gyro about the roll axis. Connect the potentiometer excitation in accordance with table I and connect pin X of table II to pin P of table I

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for sine output, P_1, to voltage ratio bridges or a ratio network, and Y of table II to pin P of table I for cosine output, P_1, to a servoed ratiometer. Connect a 100K-ohm load to each sine and cosine output in parallel with the voltage ratio bridges and with the servoed ratiometer respectively.

b. Connect the displacement gyroscope roll synchro output to a precision angle indicator.

c. Apply power and allow the system to stabilize.

d. Tilt the table to approximately the angles listed in the roll synchro, nominal column in table VI. Set the exact desired tilt angle by synchronizing to the percentage values on the voltage ratio bridges for sine and cosine outputs listed in the third and fourth columns from the left, respectively. The percentage voltage ratio for the sine and cosine outputs as read on the servoed ratiometers and the displacement gyroscope roll angle as read on the precision angle indicator shall be as specified in the roll synchro degree and minute minimum and maximum columns.

e. Repeat steps a, b, c, and d for sine and cosine outputs designated P_2, , and 15.

4.6.1.20.3 Potentiometer phasing

4.6.1.20.3.1 Pitch linear output. The pitch linear potentiometer output shall be tested as follows:

a. Mount the displacement gyroscope on a tilt table so that the table rotation axis is colinear with the gyro pitch axis. Connect the displacement gyroscope as shown in tables I and II. Connect a voltmeter and a 200K-ohm load in parallel across pin f of table II and pin P of table I.

b. Apply power and allow the system to stabilize. With the table level to within $\pm 1^{\circ}$, the output across pins f and P shall be approximately 0-V dc.

c. Rotate the tilt table in a positive pitch direction (nose up). The voltage output shall become more positive. Rotate the tilt table in a negative pitch direction (nose down). The voltage output shall become more negative.

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TABLE VI. PERCENT VOLTAGE RATIO FOR CW ROLL

Input Percent Voltage Ratio		Output Percent Voltage Ratio				Output Roll Synchro		
Nominal		Min	Max	Min	Max	Nom	Min	Max
Sine	Cosine	Sine		Cosine		Degrees, Minutes		
				9.920	10.080	0	-0, 43	0, 43
0				8.882	10.042	+5	+4, 17	+5, 43
0.872				9.579	9.739	+15	+14, 17	+15, 43
2.588						+60	+59, 13	+60, 47
	5.000	8.580	8.740			+90	+89, 17	+90, 43
	0	9.920	10.080			+120	+119, 13	+120, 47
	5.000	8.580	8.740			+135	+134, 6	+135, 54
				6.991	7.151	+180	+179, 17	+179, 17
7.071				9.920	10.080	-135	-135, 54	-134, 6
0				6.991	7.151	-120	-120, 47	-119, 13
7.071						-90	-90, 43	-89, 17
	5.000	8.580	8.740			-60	-60, 47	-59, 13
	0	9.920	10.080			-15	-15, 43	-14, 17
	5.000	8.580	8.740			-5	-5, 43	-4, 17
				9.579	9.739	0	-0, 43	0, 43
2.588				9.882	10.042	-15	-14, 17	-15, 43
0.872				9.920	10.080	0	-0, 43	0, 43
0				9.579	9.739			
2.588				9.920	10.080			
0								

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4.6.1.20.3.2 Sine-cosine roll outputs. The sine-cosine roll outputs shall be tested as follows:

a. Mount the displacement gyroscope on a tilt table so that the table rotation axis is colinear with the gyro roll axis. Connect the displacement gyroscope as per tables I and II. Connect a voltimeter and a 100K-ohm load in parallel across pin X of table II and pin P of table I for sine roll output, P_1, and across pin Y of table II and pin P of table I for cosine roll output, P_1.

b. Apply power and allow the system to stabilize. With the table level to within $\pm 1^{o}$, the sine output shall be approximately zero and the cosine output shall be approximately +15V dc.

c. Rotate the tilt table in a positive roll direction (right wing down). The voltage for the sine output shall become more positive.

d. Rotate the tilt table in a negative roll (left wing down). The voltage for the sine output shall become more negative.

e. Repeat steps a through d except check the phasing for sine and cosine outputs designated P_2, P_3, and 15.

4.6.1.21 Case leakage. The sealed case shall be tested for leaks by means a helium leak detector or other means approved by the procuring activity. The leak rate shall not be sufficient to cause more than 10 percent loss or contamination of the filling medium after 1,000 hours at a pressure differential of 12 to 14.7 psi.

4.6.1.22 Insulation resistance. A potential of 250V dc shall be applied between the following isolated pins and between the pins and the case for a period of 10 seconds. There shall be no breakdown of insulation or any other permanent damage to the displacement gyroscope as a result of this test, and the insulation resistance shall be not less than 30 megohms.

From connector pins:	To connector pins or case:
J8/N, S, U	Case
J9/A, F, W	Case
J9/P, d	Case (includes all five potentiometers)
J8/N	J9/A, F
J9/A	J9/F

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4.6.2 Electronic control amplifier

4.6.2.1 Examination of product. The amplifier shall be examined to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2.2 Infant mortality aging. The amplifier shall be operated for 48 hours. In case of any component failure or an out-of-tolerance condition, the failed component shall be replaced or the out-of-tolerance condition shall be corrected and the amplifier then subjected to the applicable individual tests prior to continuing this test. This test need not be repeated except as necessary to complete the 48-hour aging period. If additional failures or out-of-tolerance conditions occur during the aging period, they shall be corrected as specified herein. Upon completion of this test, the amplifier shall be subjected to the remaining individual tests.

4.6.2.3 Insulation resistance. A potential of 500V rms ac at a commercial frequency shall be applied between the following isolated pins and the case for a period of 5 seconds. There shall be no breakdown of insulation or any other permanent damage to the instrument as a result of this test. Insulation breakdown shall be considered as an a-c leakage impedance of less than 200,000 ohms or a current leakage of 2.5 milliamperes.

J-2001	J-2002	J-2003
B	L	F
	J	L
	A	J
	H	
	E	
	f	
	T	

4.6.2.4 Continuity. With the equipment deenergized and no system connections to the electronic control amplifier, the resistance of the circuits shall be as follows:

Connector	Pin	Resistance (Ohms)
J2001(J1)	A-B	6-18
	A-C	6-18
	B-C	6-18

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Connector	Pin	Resistance (Ohms)
J2002 (J2)	D-E	100-300
	H-F	10-22
	F-S	10-22
	S-H	10-22
	A-B	9-20
	A-C	9-20
	B-C	9-20
	J-K	4-12
	L-M	3-8
	L-N	3-8
	M-N	3-8
	U-W	Less than 1
	f-T	∞
J2003 (J3)	A-B	1-15
	K-L	∞
	K-J	Less than 1
	E-F	300-700
	D(J3)-D(J1)	Less than 1
	D(J3)-E(Test)	Less than 1
	C(J3)-d(J2)	Less than 1
	A(J3)-E(J1)	1-4
	A(J3)-i(J2)	Less than 5
	D(J3)-X(J2)	Less than 1
	A(J3)-h(J2)	Less than 1
Test Jacks	A-E	1K-6K
	D-Z(J2)	Less than 1
	B-C	25-75
	F-A(J3)	∞
	J-A(J3)	Less than 15

4.6.2.5 Automatic synchronization. Using an azimuth transmitter simulator, the precise data repeater shall be set to the 170° heading. The DT-309/AJN simulator shall be set to 0° heading. The automatic synchronization shall be started by connecting g to Y of J2002. The precise data repeater shall stop at 0° ±2° and the time for the start of automatic synchronization until autosynchronization stops shall not exceed 10 seconds. Once synchronized, oscillation shall be attempted by disturbing the flux valve simulator or azimuth simulator around null. If oscillation exists, the autosynchronization generator potentiometer shall be set to the maximum ccw position and the potentiometer rotated cw to the position which stops oscillation. When the

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synchronization cycle is ended, the annunciator needle shall be centered. Note: This test must be performed before the time-delay relay is energized. If more time is required, the automatic synchronization cycle shall be repeated by removing the jumper between g and Y of J2002 and quickly reconnecting the jumper to the same pins.

4.6.2.5.1 The time from the start of automatic synchronization until the time-delay relay is energized shall be 15 ±5 seconds.

4.6.2.5.2 The test specified in 4.6.2.5 shall be repeated with the precise data repeater set to the 190° heading.

4.6.2.5.3 The power adapter voltage and frequency measured at the test jack in the front of the electronic control amplifier shall be 26 ±2.6V 400 Hz ±5 percent ac.

4.6.2.6 Excitation voltage. The electronic control amplifier shall be given a warmup period of 15 minutes. The following excitation voltage shall meet the indicated specification:

Connection	Voltage, Volts (400 Hz)	Phase*
i (high) and h (low)	26.0 ±2.6	0° ±2°

*Reference phase is the 400-Hz line voltage applied to B (high) of J2003 and A (low) of J2003.

4.6.2.7 Power consumption. Synchronized in the slaved mode, the 400-Hz line current shall be measured in series with pin B of J2003 and shall be 200 to 350 milliamperes. The d-c line current shall be measured in series with pin C of J2003 and shall be 0.5 to 1.1 amperes.

4.6.2.8 Flux valve excitation. The electronic control amplifier shall provide 23.5 ±2.3V rms, 400 Hz for flux valve excitation.

4.6.2.9 Data transmitter nulls. With the electronic control amplifier connected to a DT-309/AJN simulator set at 0° heading, the output data transmitters shall be within 0 ±0.75°.

4.6.2.10 Slaving polarity. The system shall be synchronized on a 0° heading. With the DT-309/AJN simulator set to a 20° heading, and with a dummy torquer impedance of 2,100 ohms (2w) connected to pins a and x and b to x of J2002, the d-c current in-load across a and x shall be less than the current in-load across b to x. The difference shall be 2.3 $^{+1.1}_{-0.5}$ milliamperes.

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4.6.2.11 Data transmitter polarities. Following the test specified in 4.6.2.10, the polarities of the precise data and repeater transmitter shall be checked by turning the azimuth transmitter simulator in an increasing direction. Both repeaters shall follow in an increasing heading direction.

4.6.2.12 Slewing polarities. The polarity of the slewing motor circuit shall be checked by connecting pins R of J2002 and D of J2003. The repeaters shall move in an increasing-heading direction. Connecting pins P of J2002 and D of J2003 shall cause the repeaters to move in a decreasing heading direction.

4.6.2.13 Slaving balance. The slaving amplifier output shall be balanced by placing a short circuit across the slaving amplifier input and adjusting the balance potentiometer until a d-c null is obtained across the slaving output.

4.6.2.14 Slaving operational test. With the DT-309/AJN simulator set for 0° heading, the repeaters shall be set to approximately 0° by means of the azimuth transmitter simulator. Slaving currents shall be observed and, by means of the azimuth transmitter simulator, adjusted until the differential d-c current is at a minimum. The output of the precise data transmitter as read on a calibrated repeater shall be within ±0.75° of a 0° heading. The DT-309/AJN simulator shall be set as follows. The differential d-c current in the dummy torquer impedance of 4.6.2.10 shall meet the indicated specifications.

Input	Output	
DT-309/AJN Heading	Differential D-C Current	Polarity
10°	3.4 ma maximum	a b **
2°	1.8 ma minimum	a b **
1°	0.9 ma minimum	a b **
0°	0.12 ma maximum	
359°	0.9 ma minimum	b a *
358°	1.8 ma minimum	b a *
350°	3.4 ma maximum	b a *

* D-C current in series with a of J2002 exceeds d-c current in series with b of J2002.

** D-C current in series with b of J2002 exceeds d-c current in series with a of J2002.

4.6.2.15 Autopilot cutout. When switching controller from DG to MAG position (auto sync) (i.e., connecting g and Y of J2002), there shall be continuity between J and K of J2003 and no continuity between L and K of J2003. After the automatic synchronization cycle is completed, there shall be no continuity between J and K of J2003 and there shall be continuity between L and K of J2003.

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4.6.2.16 Manual synchronization. The precise data transmitter shall be set to 0° heading using the azimuth transmitter simulator, and the DT-309/AJN simulator shall be set to 170° heading. The transmitter shall be manually synchronized at maximum rate by turning the set control to the extreme right position. The transmitter shall move in an increasing-heading direction and the time required for it to reach the 170° heading shall not exceed 4.5 seconds. This procedure shall be repeated with the DT-309/AJN simulator set to a 190° heading by turning the set control to the extreme left. The same time limit shall apply and the transmitter shall move in a decreasing-heading direction.

4.6.2.17 Slow manual synchronization. The transmitter shall be manually synchronized at a slow rate by turning the set control to the right to the first detent position. The transmitter shall move in an increasing direction at a rate of 360° in not less than 55 seconds nor more than 240 seconds. This procedure shall be repeated except that the set control shall be turned to the left to the first detent position. The transmitter shall move in decreasing-heading direction. The same time limits shall apply.

4.6.2.18 Test jacks. Jacks on the front panel of the electronic control amplifier shall make available all pertinent input and output voltages as follows:

a. Slaving input voltage - test jacks F-E

b. Slaving output voltage - test jacks G-H

c. Azimuth input voltage - test jacks A-E

d. Azimuth output voltage - test jacks B-C

e. Power adapter output voltage - test jacks J-E

f. Latitude correction voltage - test jacks D-E.

4.6.2.19 Latitude correction voltage. The voltage measured from J1 D (low) to J2 Z (high) shall be more than 11.0V dc when the latitude correction potentiometer (R1 on front of electronic control amplifier) is set to the extreme cw position. The latitude correction potentiometer (R1) shall be set to the extreme ccw position. The voltage shall be less than 7.0V dc. A cw movement of the latitude correction potentiometer shaft (as viewed from the front panel) shall produce an increasing voltage.

4.6.2.19.1 Latitude correction voltage adjustment. The latitude correction voltage shall be adjusted to 9.5V dc by means of potentiometer R1 in the front panel. The 27.5V supply shall be raised to 30V and then lowered to 24V. The voltage previously noted shall not change by more than 0.3V.

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4.6.2.20 Azimuth servo-amplifier operation. The azimuth transmitter simulat shall be turned rapidly through an angle of approximately 45°. The indicator shall move smoothly without noticeable jumping or overshooting. There shall b no noticeable hunting or oscillation when the simulator is stationary or moving.

4.6.2.21 Azimuth followup rate. The azimuth transmitter simulator shall be turned through an angle in excess of 360°, first cw and then ccw, at a rate of not less than 7 revolutions per minute. The indicators shall rotate through 360° in 9 seconds or less for both directions.

4.6.2.22 Index error. The calibrated DT-309/AJN simulator shall be set to 0° heading and the azimuth simulator so adjusted that the differential slavi current is zero (a-b of J2002). The calibrated repeater error shall be recorded. This test shall be conducted every 15° through 360° cw and ccw. Th algebraic sum divided by the number of readings shall be less than ±0.25°. The error at any one heading shall not be more than 0.5° less the index erro

4.6.2.23 E-Z check. The calibrated indicator and precise transmitters shal be within 0.2° of each other at zero heading.

4.6.2.24 Turn cutout relay. The turn cutout relay shall be energized with 27.5V dc at terminals E and F of J2003. Circuit continuity shall be established from terminal T of J2002 to terminal f of J2002. In additi th differential current to the dummy torquer impedances shall not exceed 0.12 milliampere.

4.6.2.25 Scale error. The azimuth transmitter simulator setting shall be adjusted to 0° heading every 30° up to and including 330°; the precise transmitter shall respond to within 1° at each heading.

4.6.3 Voltage and frequency variation. Each component shall be properly connected and operated until gyroscope erection is completed. The components shall be operated for 15 minutes with an applied voltage and frequency of 103V and 320 Hz, respectively, and then for 15 minutes with an applied volta and frequency of 128V and 480 Hz, respectively. No tests shall be conducted during the 15-minute periods. The voltage and frequency shall be returned to 103V and 380 Hz, respectively, the heading circuit synchronized on a 0° head and the components subjected to the following tests with a 10 percent increa in tolerances. The system shall not be damaged by this test.

a. Gyroscope: 4.6.1.5, 4.6.1.15, 4.6.1.16, 4.6.1.20.1, and 4.6.1.20.2

b. Amplifier: 4.6.2.19.1, 4.6.2.20, and 4.6.2.21.

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4.6.3.1 Variable voltage and frequency roll servo operation. The gyroscope shall be connected as required and the roll axis shall be in a horizontal plane. At any combination of 115 ±11.5V and 400 ±80 Hz power supply, the displacement gyroscope shall be rotated about the roll axis at rates of 10° and 360° per second. During these variations, monitoring the inner roll signals using pins A to C of J-8, the voltage shall be recorded on a Sanborn recorder, or equivalent.

4.6.3.2 Low speed followup. Rotating the unit at a rate of 10° per second, the spin axis of the vertical gyro shall maintain a steady-state 90° relationship to the pitch axis within the following limits as indicated by the voltage from the inner-roll synchro. This test shall be performed in normal and special modes for right- and left-roll conditions.

Frequency (Hz)	320	400	480
Degrees deviation as read on inner roll	±0.5	±0.25	±0.25

4.6.3.3 High speed followup. Rotating the unit at a rate of 360° per second, the spin axis of the vertical gyro shall maintain a steady-state 90° relationship to the pitch axis within 15°, as indicated by a voltage from the inner roll synchro. This test shall be performed in normal and special mode and for right- and left-roll conditions. The frequencies shall be in accordance with 4.6.3.2.

4.6.3.4 Shift in null. The shift in the synchronized null as measured by one of the two precise data transmitting synchros shall not exceed $3/4^{\circ}$.

4.6.4 Low temperature operation. Each component shall be properly connected with no power applied and placed in a chamber at an ambient temperature of -54° ±2°C for a period of 4 hours. At the end of this period and while at the low temperature, the components shall be subjected to the following tests:

a. Gyroscope: 4.6.1.3, 4.6.1.4, 4.6.1.5, 4.6.1.9, 4.6.1.11, 4.6.1.15, 4.6.1.16, 4.6.1.20.1, and 4.6.1.20.2

b. Amplifier: 4.6.2.14, 4.6.2.20, 4.6.2.21, and 4.6.2.22.

4.6.4.1 Exceptions to low temperature operation test. The following exceptions shall apply to the tests specified:

4.6.4.1.1 Initial erection. Initial erection shall be accomplished within 2 minutes.

4.6.4.1.2 Final erection. Final erection shall erect the vertical gyro to within $1/2^{\circ}$ of vertical within 2 minutes and to within $1/4^{\circ}$ by the end of the additional 3 minutes.

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4.6.4.1.3 Vertical gyro drift. Maximum free drift shall not exceed 2.5°.

4.6.4.1.4 Random wander. The drift of the directional gyro for each 1/2-hour period (through a 2-hour test) shall not exceed 1.5°.

4.6.4.1.5 Latitude correction. The error shall not exceed ±3-1/2° per hour.

4.6.4.1.6 Scale error. Scale error shall not exceed 1/2°.

4.6.4.1.7 Dynamic accuracy. The error shall not exceed ±2°.

4.6.4.1.8 Slaving operational test tolerance. The differential d-c current shall have a 10-percent increase in tolerance.

4.6.4.1.9 Potentiometer outputs. The tests specified in 4.6.1.20.1 and 4.6.1.20.2 shall be conducted except the tolerances shall be increased by 10 percent.

4.6.5 High temperature operation. Each component shall be properly connected, placed in a chamber at a temperature of 71° ±2°C, and operated for a 4-hour period. At the end of this period and while at the high temperature, the components shall be subjected to the following tests:

a. Gyroscope: 4.6.1.3, 4.6.1.4, 4.6.1.5, 4.6.1.9, 4.6.1.11, 4.6.1.13, 4.6.1.15, 4.6.1.16, 4.6.1.20.1, and 4.6.1.20.2

b. Amplifier: 4.6.2.14, 4.6.2.20, 4.6.2.21, and 4.6.2.22.

4.6.5.1 Exceptions to high temperature operation test. The following exceptions shall apply to the tests specified:

4.6.5.1.1 Vertical gyro drift. Maximum free drift shall not exceed 2-1/2°.

4.6.5.1.2 Random wander. The directional gyro drift for each 1/2-hour period (through a 2-hour test) shall not exceed 1.5°.

4.6.5.1.3 Latitude correction. The error shall not exceed ±3-1/2° per hour.

4.6.5.1.4 Scale error. The scale error shall not exceed ±1/2°.

4.6.5.1.5 Dynamic accuracy. The error shall not exceed ±2°.

4.6.5.1.6 Slaving operational test tolerance. The differential d-c current shall have a 10-percent increase in tolerance.

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4.6.5.1.7 Potentiometer outputs. The tests specified in 4.6.1.20.1 and 4.6.1.20.2 shall be performed except the tolerances shall be increased by 10 percent.

4.6.6 Dielectric strength. A potential of 500V dc shall be applied between isolated pins and between pins and the case of each component for a period of 10 seconds. There shall be no insulation breakdown. This test shall be performed on the displacement gyroscope before sealing.

4.6.7 Vibration error. Each component shall be properly connected for normal operation in the slaved mode, power applied, and subjected to a vibration error test. During this test, the input vibration to the equipment shall coincide with a power spectral density of $0.015g^2$ per Hz for each frequency from 20 to 2,000 Hz. To accomplish this, the loading effect of the unit under test shall be eliminated by proper load compensation. This random vibration shall be applied to the equipment along the three mutually perpendicular axes of the component. The signal source shall be either a prepared magnetic tape or a noise generator. The duration shall be from 10 to 15 minutes.

4.6.7.1 Pitch, roll, and azimuth signals. The pitch, roll, and azimuth signals from the displacement gyroscope shall not vary by more than $1/2^{\circ}$ and shall return to the initial position when the vibration ceases. The outputs of the data transmitters shall vary not more than $3/4^{\circ}$ from the initial position and shall return to the initial position when the vibration ceases.

4.6.7.2 Potentiometer output signals. The potentiometer output signals (linear pitch, sine roll, and cosine roll) at null shall be as follows:

4.6.7.2.1 Pitch (linear). With the voltage ratio percentage set at 50 percent, the pitch angle readout shall be $0^{\circ} \pm 1^{\circ}$.

4.6.7.2.2 Roll (sine and cosine). With the sine output voltage ratio percentage set at 50 percent, the cosine output ratio percentage as read on the servoed ratiometer shall be 50 ±1 percent and the roll angle readout shall be $0^{\circ} \pm 1^{\circ}$.

4.6.8 D-C voltage variation. The electronic control amplifier shall be properly connected and power applied. The system shall then be operated with an applied d-c voltage of 24V. Using the azimuth transmitter simulator, the precise data repeater shall be set to a 170° heading and the DT-309 simulator shall be set to 0° heading. The automatic synchronization shall be started by connecting g to Y of J2002. The precise data repeater shall stop at $0^{\circ} \pm 2^{\circ}$ and the time from the start of automatic synchronization until the time of stopping shall not exceed 14 seconds. The d-c voltage shall be raised to 30V and the test repeated.

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4.6.9 Electromagnetic interference and susceptibility. The electromagnetic interference and susceptibility test shall be conducted in accordance with MIL-STD-462, class A1, test methods CE03, CE04, CS01, CS02, CS06, RE02, RS02, and RS03. The interference test shall be conducted over a frequency range of 150 KHz to 400 MHz and the susceptibility test shall be conducted over a frequency range of 150 KHz to 1,000 MHz. During this test, the gyroscope shall be connected properly and shall be operated on a 15° roll, pitch, and yaw mechanism.

4.6.10 System compatibility tests. The systems specified in 4.5.2.3 shall be subjected to the following tests:

4.6.10.1 Azimuth servo operation. The system shall be connected properly and operated. The displacement gyroscope shall be rotated rapidly through an angle of approximately 45° about its azimuth axis. The azimuth servo shall move smoothly without noticeable jumping or overshooting. There shall be no noticeable hunting or oscillating of the azimuth servo when the displacement reference is stationary or moving.

4.6.10.2 Azimuth followup rate. The system shall be connected properly and operated. The displacement gyroscope shall be turned through an angle in excess of 360°, first cw and then ccw at a rate of not less than 7 rpm. The azimuth servo shall rotate through 360° in 9 seconds or less for both cw and ccw rotation.

4.6.10.3 Azimuth slaving rate and cutout. The system shall be set into operation in the MAG mode and synchronized with the magnetic detector on a north heading. The magnetic detector shall then be turned approximately 30° cw and the displacement through which the gyro shall be torqued shall be between 10° and 20° for a 10-minute period with proper correction for the earth's rotation. External deslaving connection shall be made in accordance with 3.5.4.4 and torquing on the directional gyro shall cease; it shall continue when the connections are broken. The function selector switch shall be turned to the DG position and the directional gyro shall cease to be torqued. This test shall be repeated with the detector turned initially approximately 30° in a ccw direction. The same tolerance shall apply.

4.6.10.4 Polarity. The system shall be set into operation and synchronized manually on a 0° heading. The magnetic detector shall be turned 30° in a ccw direction (numerically decreasing heading). The direction of rotation of the repeater indicator pointer shall be in a numerically decreasing direction. The annunciator shall point to the operator's right in the sector designated by the minus sign. A ccw rotation in the direction indicated for minus on the synchronizer control shall cause the annunciator to move to the left and synchronize with the magnetic detector when the annunciator reaches its vertical

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position. The procedure shall then be repeated, turning the magnetic transmitter in a cw direction. The direction of rotation of the repeater indicator pointer shall be in a cw direction and the annunciator shall point to the operator's left in the plus sector. A cw rotation in the direction designated by a plus sign on the synchronizer control shall cause the annunciator to move to the right and the system to synchronize when the vertical position is reached.

4.6.10.5 Static settling. The system shall be set in operation and to a synchronized null. The displacement reference shall be rotated 10° from the null position. The output of the precise data transmitter shall be read after a period of 15 minutes has elapsed. This procedure shall be followed for both sides of null. The difference between the two final settling positions shall be not greater than ±1/4°.

4.6.10.6 Dynamic accuracy. The system shall be set in operation and to a synchronized null. The magnetic detector shall be oscillated through ±5° either side of the null at 6 cpm; at 1-minute intervals, high and low readings shall be made. Readings shall continue until three consecutive 1-minute readings are within 0.1° of each other; this shall occur in less than 45 minutes from the start of the test. The difference between the initial null value and the average end value (determined by averaging the high and low readings of the last three consecutive readings) shall be no greater than 1.5°.

4.6.10.7 Scale error. A magnetic detector with known error, or an approved equivalent simulator, mounted on a suitable turntable in an area having magnetic quality which will permit the performance of this test, shall be connected to the system. The system shall be set into operation in the MAG mode and synchronized to a magnetic north signal. The detector shall be rotated in 15° increments through 360° cw and the system manually set to a synchronized null on each heading. A unit containing a synchro of known calibration mounted in an accurate positioning device and a null reading meter, or their equivalents, shall be connected electrically to the precise data transmitter for accurate determination of compass heading. These readings shall be recorded for each position of the magnetic detector. The test shall be repeated for each successive 15° heading, from 0° with a ccw rotation of the magnetic detector. The deviation thus determined by both rotations shall be added algebraically and the result divided by the number of readings. The result shall be the index error and shall not exceed ±0.25°. The readings shall not differ from the correct heading, less the index error, by more than 0.5°. The difference of like readings as a result of ccw rotation shall not exceed ±0.5°. The electrical zero position for the indicator data transmitter shall be the same as that for the precise data transmitter.

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4.6.10.8 Synchronization

4.6.10.8.1 Power application synchronization. The system shall be set into operation in the magnetic slaved mode and synchronized with the magnetic detector on a north heading. The electrical power shall then be removed, the magnetic detector rotated cw 170°, and the electrical power reapplied to the system. The system shall synchronize automatically to within 2° of the 170° heading in not more than 10 seconds from application of power, if initial erection is not recycled, or within the time period of the initial erection cycle if power is removed long enough to cause recycle.

4.6.10.8.2 The procedures of 4.6.10.8.1 shall be repeated in order, except in the opposite direction of rotation. The same tolerances shall be met.

4.6.10.8.3 Switch-over synchronization. The procedure of 4.6.10.8.1 shall be repeated except that the function selector switch shall be placed momentarily in the DG position and then returned to the MAG position in lieu of removing the electrical power from the system. The compass shall synchronize to within 2° of the 170° heading within 10 seconds.

4.6.10.8.4 The procedures of 4.6.10.8.3 shall be repeated in order, except in the opposite direction of rotation. The same tolerances shall be met.

4.6.10.9 Heading set slew. The system shall be set into operation in the DG mode. One of the data transmitter synchros and the indicator differential synchro shall be monitored. The synchronizer and heading slew control shall be pressed and rotated cw and the signal output shall indicate a cw heading change. The maximum rate shall be not less than 34° per second. The minimum rate shall be 1.5° to 6.5° per second. This procedure shall be repeated except the rotation shall be in the ccw direction. The same performance requirements shall be met.

4.6.10.10 System voltage and frequency variation. The system shall be tested in accordance with 4.6.3 after which it shall meet the tests specified in 4.6.10.1 and 4.6.10.2 except during the test specified in 4.6.10.2, the tolerance shall be increased 10 percent.

4.6.10.11 System low temperature operation. The system shall be tested in accordance with 4.6.4 after which it shall meet the tests specified in 4.6.10.1, 4.6.10.2, 4.6.10.3, 4.6.10.6, and 4.6.10.7. Exceptions shall be as listed in 4.6.4.

4.6.10.12 System high temperature operation. The system shall be subjected to the test specified in 4.6.5 after which it shall meet the tests specified in 4.6.10.1, 4.6.10.2, and 4.6.10.3.

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4.6.11 Environmental tests. The following tests, except 4.6.11.5.1 and 4.6.11.6, shall be conducted in accordance with the applicable procedures of MIL-STD-810 and as specified herein:

4.6.11.1 Temperature-altitude. The components shall be properly connected with no power applied and subjected to a temperature-altitude test in accordance with method 504, procedure I, equipment class 2, except the pressure in the chamber shall be the equivalent of 80,000 ±500 feet for steps 3, 11, and 12. Power shall be applied and the time required for initial erection of the gyros shall not exceed 5 minutes. The displacement gyroscope shall be rotated through 360° about the pitch, roll, and azimuth axes and the components shall perform their respective functions without binding or erratic movement. After the components have returned to room temperature, they shall meet the tests specified in 4.6.1.3 through 4.6.1.5, 4.6.1.11, 4.6.1.14 through 4.6.1.16, 4.6.1.20.1, 4.6.1.20.2, 4.6.2.14, 4.6.2.20 through 4.6.2.22, 4.6.10.2, 4.6.10.3, and 4.6.10.7, as applicable.

4.6.11.2 High temperature exposure. The components shall be properly connected with no power applied and subjected to a high temperature exposure test in accordance with method 501, procedure I. The maximum operating temperature shall be 71°C. At the end of this test and while still at the high temperature, power shall be applied. The gyro transmitter shall be rotated through 360° about the pitch, roll, and azimuth axes. The components shall perform their respective functions without binding or erratic movement. After the components have returned to room temperature, they shall meet the tests specified in 4.6.1.3 through 4.6.1.5, 4.6.1.9, 4.6.1.11, 4.6.1.14 through 4.6.1.16, 4.6.1.20.1, 4.6.1.20.2, 4.6.2.14, 4.6.2.20, 4.6.2.21, 4.6.10.2, and 4.6.10.6, as applicable.

4.6.11.3 Low temperature exposure. The components shall be properly connected with no power applied and subjected to a low temperature exposure test in accordance with method 502, procedure I. The exposure period shall be 48 hours, the exposure temperature shall be -65°C, and the operating temperature shall be -54°C. At the completion of this test, the temperature shall be raised to -54°C, stabilized, and then power shall be applied. The gyro shall be rotated through 360° about the pitch, roll, and azimuth axes. The components shall perform their respective functions without binding or erratic movement. After the components have returned to room temperature, they shall meet the tests specified in 4.6.1.3 through 4.6.1.5, 4.6.1.9, 4.6.1.11, 4.6.1.14 through 4.6.1.16, 4.6.1.20.1, 4.6.1.20.2, 4.6.2.14, 4.6.2.20, 4.6.2.21, 4.6.10.2, and 4.6.10.6, as applicable.

4.6.11.4 Temperature shock. The components shall be subjected to a temperature shock test in accordance with method 503. After the components have returned to room temperature, the connectors and seals shall be examined and

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there shall be no evidence of cracked connectors or of leaks in the seals. The components shall then meet the tests specified in 4.6.1.11, 4.6.1.15, 4.6.1.16, 4.6.1.20.1, 4.6.1.20.2, 4.6.2.17, and 4.6.2.20, as applicable.

4.6.11.5 Vibration failure

4.6.11.5.1 Displacement gyroscope. With the displacement gyroscope connected and operating, vibration shall be applied along each of the three mutually perpendicular axes in accordance with the acceleration schedule shown on figure 6 and the frequency cycling period shown on figure 7 for the following vibration time periods per axis: 180 minutes at room temperature; 60 minutes at +71°C; and 60 minutes at -54°C. At the end of this test, the displacement gyroscope shall meet the tests specified in 4.6.1.3 through 4.6.1.13, 4.6.1.15, 4.6.1.16, 4.6.1.20.1, and 4.6.1.20.2. All resonant points shall be noted and reported.

4.6.11.5.2 Electronic control amplifier. With the electronic control amplifie connected and operating, vibration shall be applied in accordance with method 5 equipment category (b), procedure I (with vibration isolators, part 1 only), except the curve shown on figure 8 herein shall apply. The amplifier shall be shock mounted during this test and shall be operated during the entire test. At the end of this test, the amplifier shall meet the individual tests.

4.6.11.6 Acceleration. The components shall be subjected to an accelera n of 20g for a period of 1 minute, first along the vertical axis and then along each of two axes that are perpendicular to the vertical axis and to each other. At the end of this test, the components shall meet their respective individual tests.

4.6.11.7 Shock. The components shall be subjected to a shock test in accordar with method 516, procedures I and III, for flight vehicle equipment, with power applied. The shock pulse configuration for the displacement gyroscope shall be in accordance with figure 516.1-1, amplitude (a) duration (c), and for the electronic control amplifier shall be in accordance with figure 516.1-2, amplitude (a) and duration (c). At the completion of procedure I testing, the components shall meet their respective individual tests.

4.6.11.8 Humidity. The components shall be subjected to a humidity test in accordance with method 507, procedure I. At the conclusion of this test, there shall be no evidence of corrosion or rusting of external parts or moisture inside the hermetically sealed case of the displacement gyroscope. The amplifi shall then meet the individual tests and the displacement gyroscope shall meet the case leakage test.

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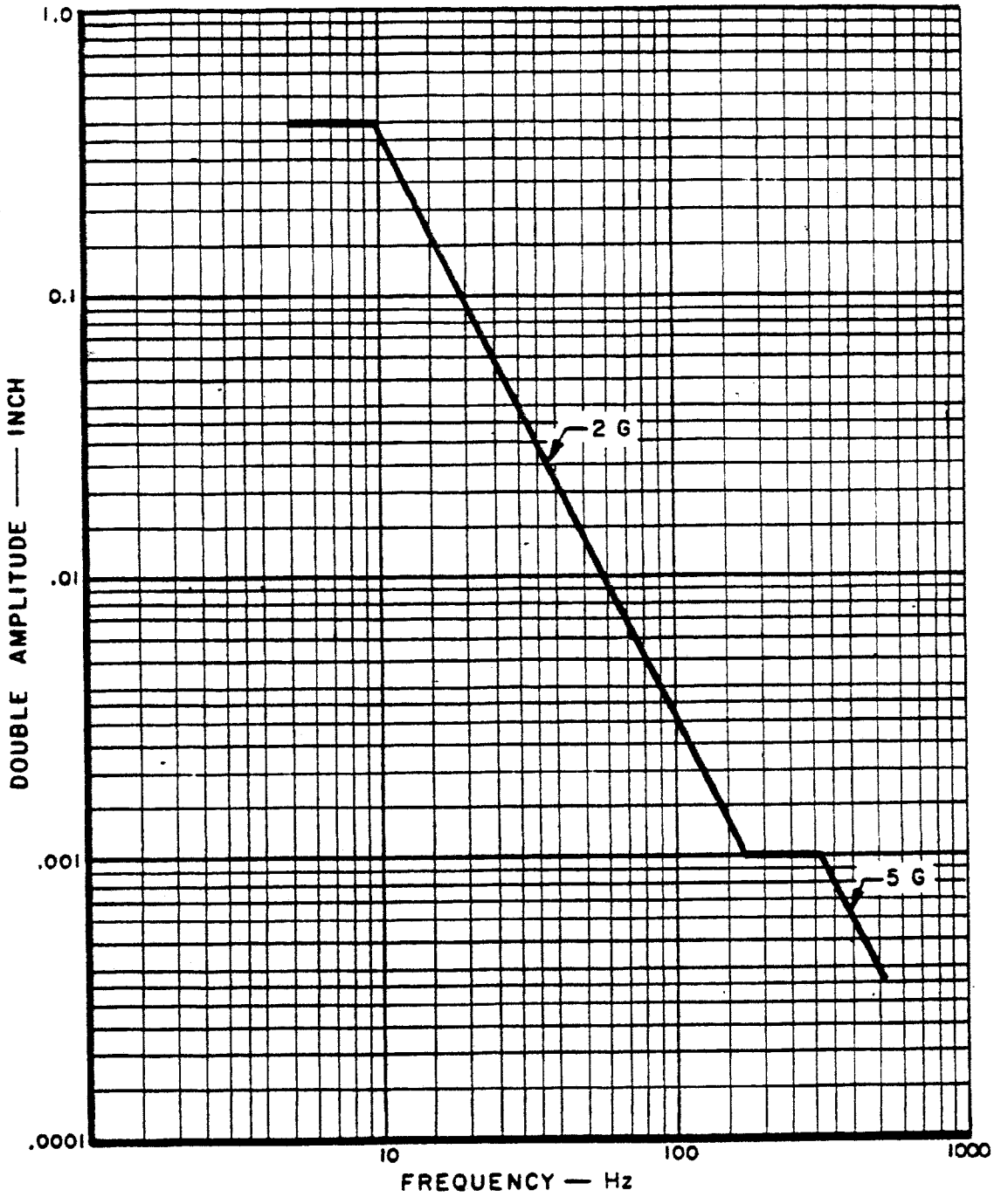


FIGURE 6. Vibration Profile

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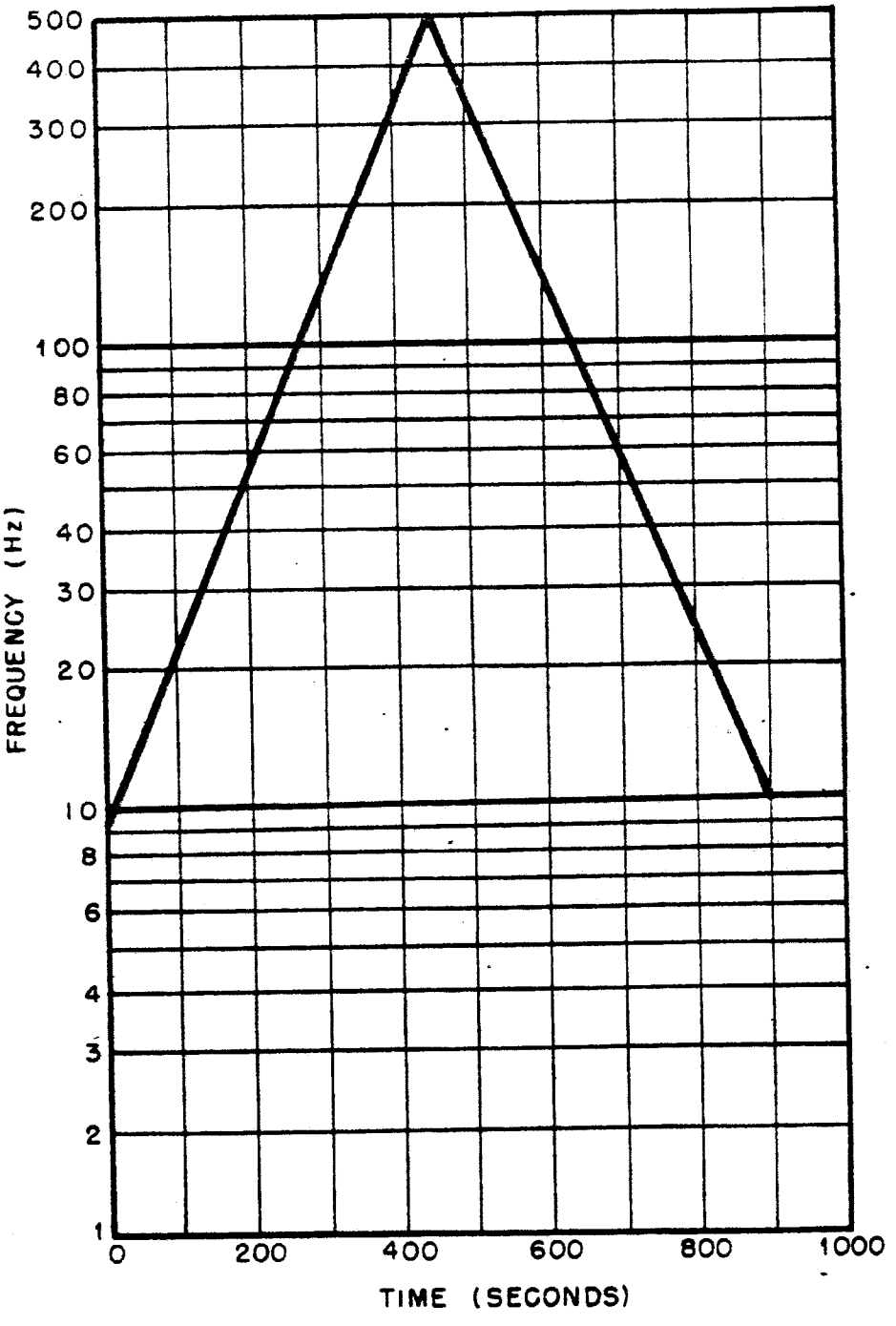


FIGURE 7. Vibration Cycling Test Schedule

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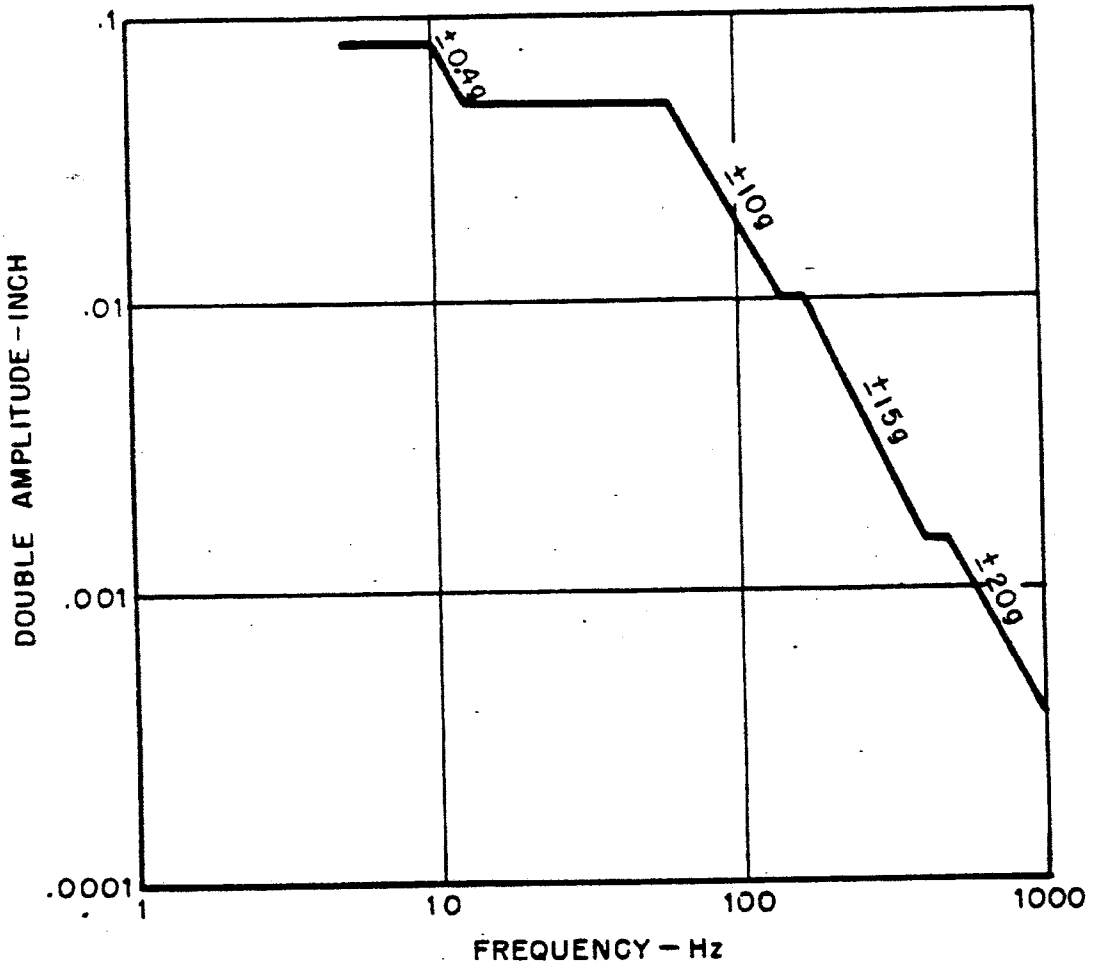


FIGURE 8. Vibration Test Curve for the Electronic Control Amplifier

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4.6.11.9 Fungus. The components shall be subjected to a fungus test in accordance with method 508. Upon completion of this test, there shall be no evidence of deterioration, corrosion, or fungus growth on the external surface of the components. The displacement gyroscope shall also meet the case leakage test.

4.6.11.10 Rain. The amplifier shall be subjected to a rain test in accordance with method 506. At the conclusion of this test, there shall be no evidence of water penetration or damage to the amplifier. The amplifier shall then meet the tests specified in 4.6.2.19 through 4.6.2.21 except that the test period for each shall be 10 minutes.

4.6.11.11 Salt fog. The salt fog test shall be conducted in accordance with method 509. After completion of this test, corrosion shall not be such as to shorten the operational life of the components. The components shall then meet the tests specified in 4.6.2.19 through 4.6.2.21, 4.6.10.3, and 4.6.10.8. The displacement gyroscope shall meet the case leakage test.

4.6.11.12 Dust. The amplifier shall be subjected to a dust test in accordance with method 510. Upon completion of this test, the components shall be examined and operated. There shall be no evidence of dust penetration or damage to the component. The components shall then meet the tests specified in 4.6.2.19 through 4.6.2.21 except the period for each setting shall be 10 minutes.

4.6.11.13 Explosion test. The amplifier shall be subjected to an explosion test in accordance with method 511, procedure I.

4.6.12 Reliability. Prior to starting the reliability test, the components specified in 4.5.2.4 shall be subjected to the individual tests after which they shall each be cycled for 1,236 hours of power-on time. During the 48-hour cycles (see figure 9) the displacement gyroscope shall be oscillated (during the power-on time) on a 15° total-movement roll, pitch, and yaw mechanism having 5 to 7 oscillations per minute.

4.6.12.1 Temperature extremes. Every 48 hours the components shall be temperature cycled over the entire range in four 12-hour periods. The +50°C and -35°C temperatures shall be maintained for 5-1/2 hours each. The time for temperature transition shall be 30 minutes.

4.6.12.2 Vibration. The components shall be vibrated at an acceleration of at least 0.2g at a frequency of 25 ±5 Hz during the last 2 hours of the cold-soak during the heat-up, and during the hot-soak time spans of the second and fourth periods.

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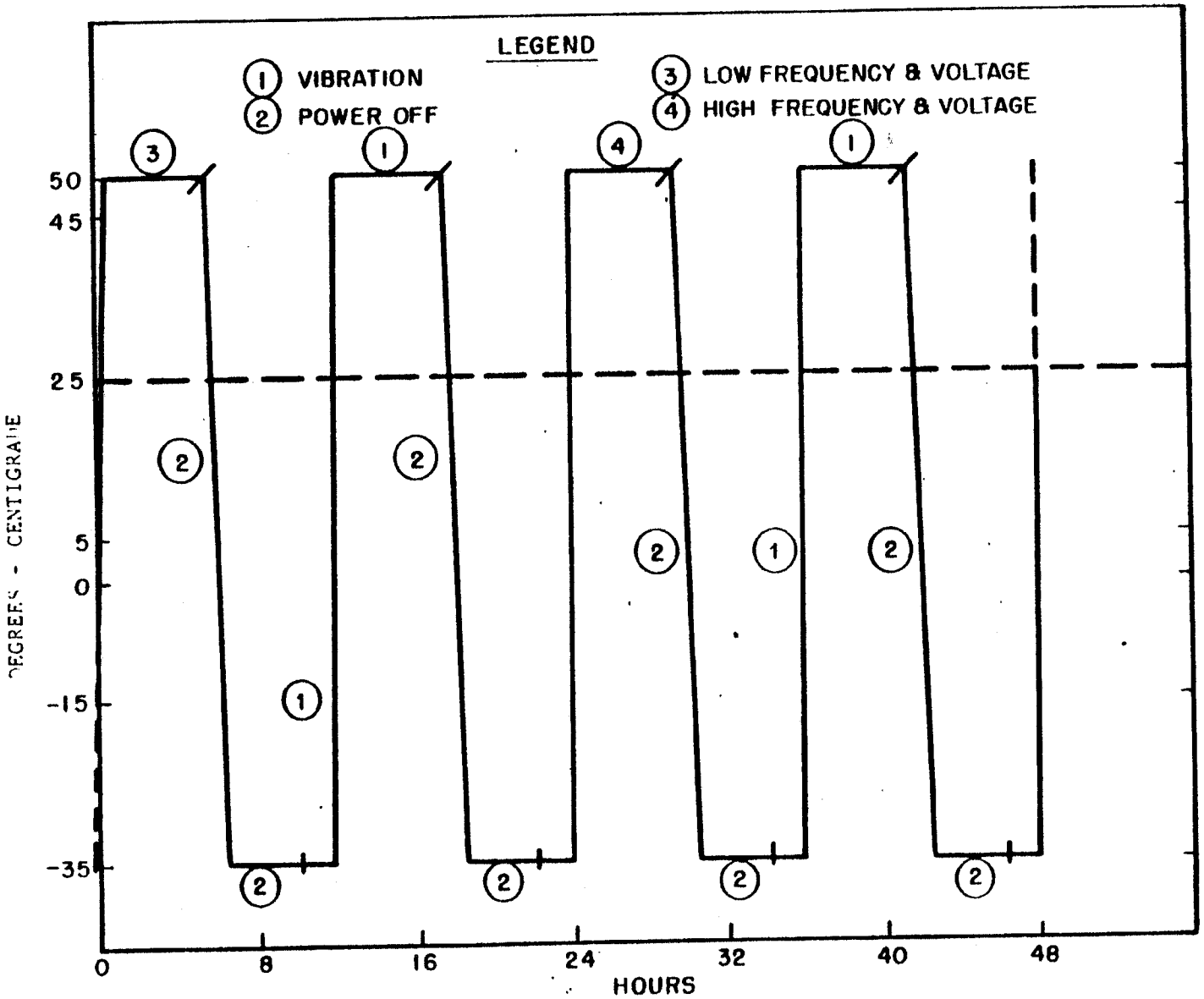


FIGURE 9. Reliability Cycling

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4.6.12.3 Low frequency and voltage. The frequency and voltage shall be set at 320 Hz and 105V respectively for 5-1/2 hours during the high-temperature soak of the first period.

4.6.12.4 High frequency and voltage. The frequency and voltage shall be set at 480 Hz and 127V respectively for 5-1/2 hours during the high-temperature soak of the third period.

4.6.12.5 Power off. Power to the components shall be turned off during the 30-minute cool-down spans and the first 3-1/2 hours of the cold-soak spans of each period. During the power-off time, the scorsby mechanism shall not be operated.

4.6.12.6 Test records, reports, and malfunctions. All out-of-tolerance conditions shall be considered as failures. A record shall be made of the failure details and the equipment shall be repaired. The extent of the repair made shall be recorded and testing shall be resumed from the point of interruption. Test records, test reports, and failure summary analyses shall be prepared in accordance with MIL-STD-781 and shall be submitted to the procuring activity

4.6.12.7 Test schedule. Each component or system, as applicable, shall be subjected to the following tests, in the order listed, at the end of every 300 hours of testing per component:

Gyroscope: 4.6.1.3 through 4.6.1.5, 4.6.1.9, 4.6.1.11, 4.6.1.13, 4.6.1.15, and 4.6.1.16. All tests specified in 4.6.1.20 shall be performed prior to and at the end of the reliability test. At the 300-hour intervals during the reliability test only, the tests specified in 4.6.1.20.1 and 4.6.1.20.2 shall be conducted in addition to the tests specified above only at $+15^{\circ}$, $+10^{\circ}$, $+5^{\circ}$, 0°, -5°, -10°, and -15°.

Amplifier: 4.6.2.14 and 4.6.2.20 through 4.6.2.22.

4.6.12.7.1 At the end of the 1,236 hours of operation per component, the individual and sampling plan C tests (except 4.5.2.3.1) shall be performed. This test schedule shall not be considered a test limitation; i.e., the contractor may perform tests oftener to assure proper operation of the components or system.

4.6.12.8 Tolerances and failures. A 50-percent increase in tolerances will b permitted after 600 operating hours (per component) of reliability testing and a 100-percent increase in tolerances will be permitted after completion of 1,236 hours (per component) of reliability testing. Out-of-tolerance calibration-type conditions will be considered failures only if they occur in two successive test cycles whether the tests are performed at a 300-hour multiple or oftener. Except as specified herein, a failure shall be as defined in 6.3.2.

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4.6.12.9 MTBF. Each component shall have a minimum acceptable MTBF of 570 hours at 90 percent confidence level with not more than eight failures when tested as specified in 4.6.12. The 90 percent MTBF shall be derived from an estimated MTBF equation defined as:

$$MTBF_{est} = \frac{T_T}{f + 1}$$

where: T_T = total test time
f = number of failures

4.6.12.10 Valid time. Component operating time accumulated during tests preliminary to reliability cycling and during the periodic testing specified in 4.6.12.7 shall not be included in the required 1,236 hours of cycling. The TTM may be used to record the test time.

4.6.13 Maintainability demonstration. The contractor shall prepare a maintainability demonstration test plan in accordance with MIL-STD-471, appendix B, test method 2, and submit it to the procuring activity for approval.

4.6.14 Compatibility (see 6.1.2). Prototype installation and flight compatibility tests shall consist of at least the following: Initial erection, final erection, erection rate, erection cutout, manual fast erection, DG and VG drift, power interruption, loop test, potentiometer outputs and sampling plan C tests as applicable. As many of these tests as possible shall be conducted on the ground prior to aircraft takeoff. The rest of the tests shall be conducted in the air during straight and level flight, coordinated turns, and acrobatic maneuvers.

4.7 Inspection and testing of preservation, packaging, and packing. The inspection and testing of preservation, packaging, and packing shall be in accordance with section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Control assemblies shall be packaged and packed in accordance with MIL-STD-794. Nonhermetically sealed components shall be preserved in accordance with MIL-P-116, method II, and hermetically sealed components shall be preserved in accordance with method III. Electrical connectors on hermetically sealed components shall be provided with water-vaporproof caps and gaskets.

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5.2 Marking of shipment. Interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129. The shipment marking nomenclature shall be as follows, as applicable:

CONTROL ASSEMBLY, GYROSCOPE, ATTITUDE-HEADING A/A24G-42

GYROSCOPE, DISPLACEMENT, SBK-16/A24G-42

AMPLIFIER, ELECTRONIC CONTROL ASK-8A/A24G-5

5.2.1 Reinspection date. The reinspection date markings shall be as specifie by the procuring activity.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. When sampling plan B tests will not be conducted (see 4.5.2.2)

c. When sampling plan D tests will not be conducted (see 4.5.2.4)

d. Levels of packaging and packing required and reinspection date marking.

6.1.1 Report. A report of all failures occurring during preproduction or quality conformance testing will be submitted to the procuring activity within 2 working days.

6.1.2 Changes or modifications. Changes or modifications resulting from the test specified in 4.6.14 will be accomplished by the contractor on all production systems at no extra cost to the government. Any system delivered prior to completion of satisfactory flight tests will be reworked and updated as necessary at no additional cost to the government. If changes are class I, all preproduction tests will be rerun on the reworked units.

6.2 Test report. The contractor will submit three copies of the report specified in 4.4.2 to the procuring activity for approval.

6.3 Definitions

6.3.1 Hermetic seal. A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic, or both, materials. A hermetic seal is not intended to include seals accomplished by gaskets.

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6.3.2 Failure. Where a condition is specified as a requirement, a failure will be considered as any electrical or mechanical condition which fails to comply with the requirements specified herein.

Custodian:
Air Force - 11

Review activity:
Air Force - 71

Preparing activity:
Air Force - 11

Project No. 6610-F197

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Inclu Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION (Y' 'DD)

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